

Solidification mechanism – Technology Significance

- **Primary processing** - Processes involving casting of molten metals into ingots or semi-finished useful shapes such as slabs.
- **Secondary processing** - Processes such as rolling, extrusion, etc. used to process ingots or slabs and other semi-finished shapes.

Solidification mechanism – Nucleation

- **Nucleation**

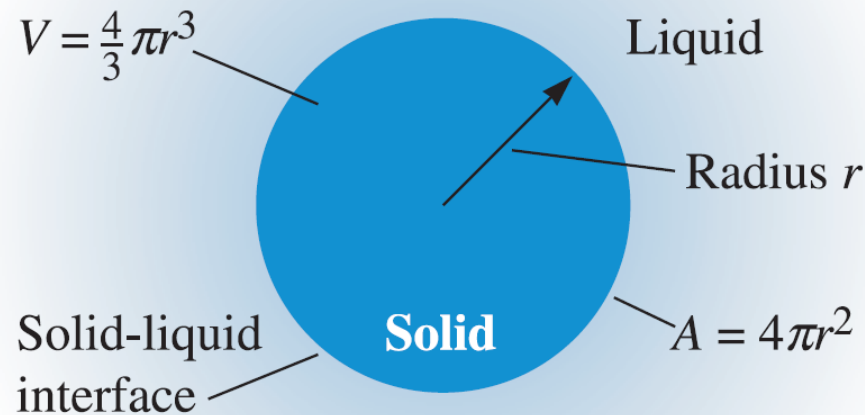
- In the context of solidification, the term nucleation refers to the formation of the first nanocrystallites from molten material.
- The physical process by which a new phase is produced in a material. (initial formation of one phase from another phase)
- nuclei (seeds) act as templates on which crystals grow
- for nucleus to form rate of addition of atoms to nucleus must be faster than rate of loss
- once nucleated, growth proceeds until equilibrium is attained

- **Condition**

- Critical radius (r^*) - The minimum size that must be formed by atoms clustering together in the liquid before the solid particle is stable and begins to grow.
- Undercooling - The temperature to which the liquid metal must cool below the equilibrium freezing temperature before nucleation occurs.

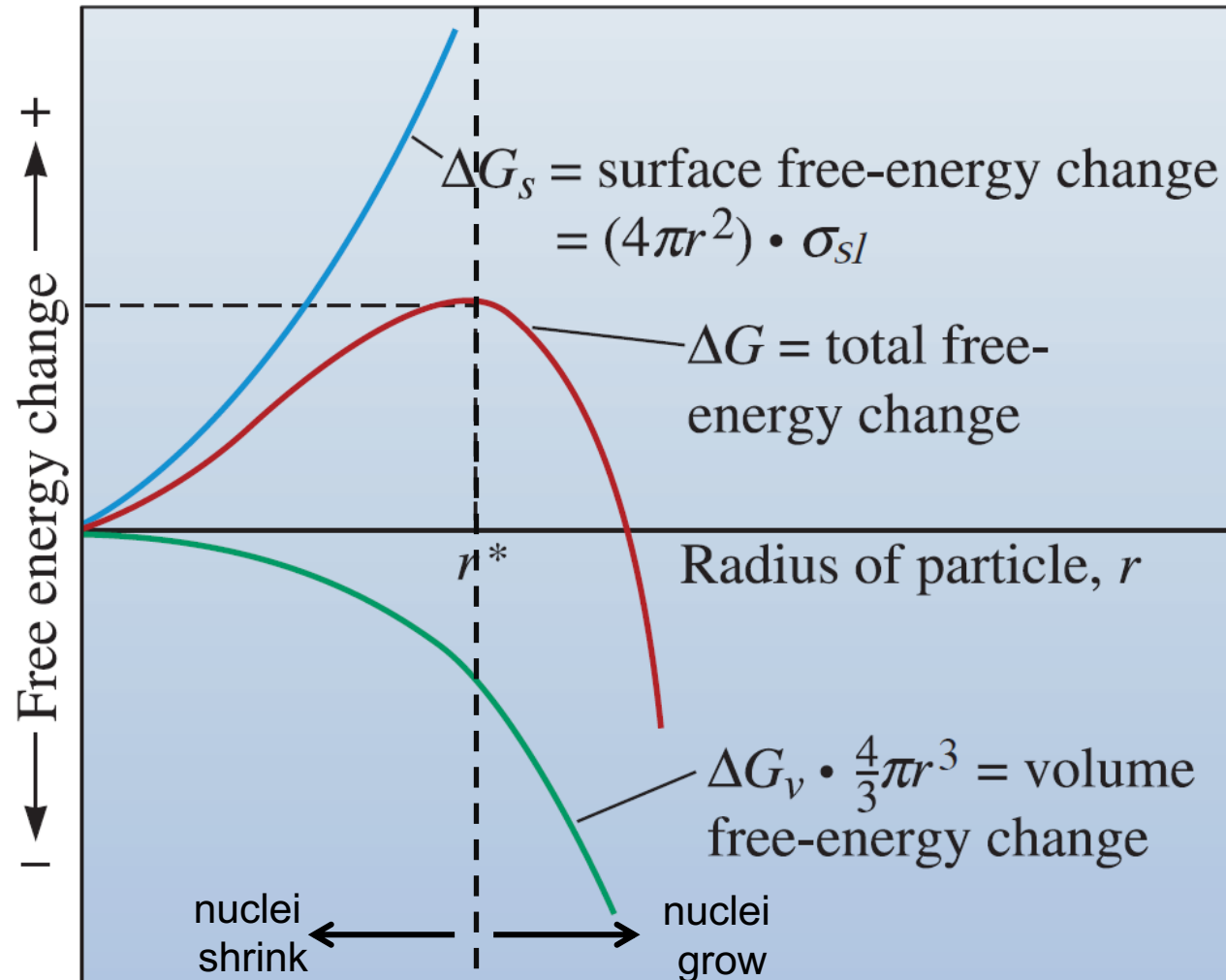
Solidification mechanism - Critical radius

An interface is created when a solid forms from the liquid



We expect a material to solidify when the liquid cools to just below its freezing (or melting) temperature, because the energy associated with the crystalline structure of the solid is then less than the energy of the liquid.

Solidification mechanism - Critical radius



When the solid forms, however, a solid-liquid interface is created. A surface free energy (σ_{sl}) is associated with this interface

$$\Delta G = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \sigma_{sl}$$

This energy difference between the liquid and the solid is the free energy per unit volume G_v and is the driving force for solidification.

The total free energy of the solid-liquid system changes with the size of the solid.

The solid is an **embryo** if its radius is less than the critical radius, and is a **nucleus** if its radius is greater than the critical radius

Solidification mechanism - Undercooling

- At the thermodynamic melting or freezing temperatures, the probability of forming stable, sustainable nuclei is extremely small. Therefore, solidification does not begin at the thermodynamic melting or freezing temperature.
- If the temperature continues to decrease below the equilibrium freezing temperature, the liquid phase that should have transformed into a solid becomes thermodynamically increasingly unstable.
- Because the temperature of the liquid is below the equilibrium freezing temperature, the liquid is considered undercooled.
- The **undercooling (T)** is the difference between the equilibrium freezing temperature and the actual temperature of the liquid.
- As the extent of undercooling increases, the thermodynamic driving force for the formation of a solid phase from the liquid overtakes the resistance to create a solid-liquid interface.

Solidification mechanism – Nucleation

- **Homogeneous nucleation** - Formation of a critically sized solid from the liquid by the clustering together of a large number of atoms at a high undercooling (without an external interface).
- **Heterogeneous nucleation** - Formation of a critically sized solid from the liquid on an impurity surface.

Homogeneous Nucleation

- As liquid cools to temperatures below the equilibrium freezing temperature, two factors combine to favor nucleation.
- First, since atoms are losing their thermal energy, the probability of forming clusters to form larger embryos increases.
- Second, the larger volume free energy difference between the liquid and the solid reduces the critical size (r^*) of the nucleus.
- Homogeneous nucleation occurs when the undercooling becomes large enough to cause the formation of a stable nucleus.

Solidification mechanism

Experimentally observed values for homogeneous nucleation

TABLE 9-1 ■ *Values for freezing temperature, latent heat of fusion, surface energy, and maximum undercooling for selected materials*

	Freezing Temperature (T_m)	Heat of Fusion (ΔH_f)	Solid-Liquid Interfacial Energy (σ_{sl})	Typical Undercooling for Homogeneous Nucleation (ΔT)
Material	(°C)	(J/cm ³)	(J/cm ²)	(°C)
Ga	30	488	56×10^{-7}	76
Bi	271	543	54×10^{-7}	90
Pb	327	237	33×10^{-7}	80
Ag	962	965	126×10^{-7}	250
Cu	1085	1628	177×10^{-7}	236
Ni	1453	2756	255×10^{-7}	480
Fe	1538	1737	204×10^{-7}	420
NaCl	801			169
CsCl	645			152
H ₂ O	0			40

Homogeneous Nucleation

The size of the critical radius r^* for homogeneous nucleation is given by

$$r^* = \frac{2\sigma_{sl}T_m}{\Delta H_f\Delta T}$$

where ΔH_f is the latent heat of fusion per unit volume,

T_m is the equilibrium solidification temperature in kelvin, and

$\Delta T = (T_m - T)$ is the undercooling when the liquid temperature is T .

- *The latent heat of fusion represents the heat given up during the liquid-to-solid transformation.*
- *As the undercooling increases, the critical radius required for nucleation decreases.*

Calculation of Critical Radius for the Solidification of Copper

Calculate the size of the critical radius and the number of atoms in the critical nucleus when solid copper forms by homogeneous nucleation. Comment on the size of the nucleus and assumptions we made while deriving the equation for the radius of the nucleus.

From Table 9-1 for Cu:

$$\Delta T = 236^{\circ}\text{C} \quad T_m = 1085 + 273 = 1358 \text{ K}$$

$$\Delta H_f = 1628 \text{ J/cm}^3$$

$$\sigma_{sl} = 177 \times 10^{-7} \text{ J/cm}^2$$

Thus, r^* is given by

$$r^* = \frac{2\sigma_{sl}T_m}{\Delta H_f\Delta T} = \frac{(2)(177 \times 10^{-7})(1358)}{(1628)(236)} = 12.51 \times 10^{-8} \text{ cm}$$

Calculation of Critical Radius for the Solidification of Copper

The lattice parameter for FCC copper is $a_0 = 0.3615 \text{ nm} = 3.615 \times 10^{-8} \text{ cm}$. Thus, the unit cell volume is given by

$$V_{\text{unit cell}} = (a_0)^3 = (3.615 \times 10^{-8})^3 = 47.24 \times 10^{-24} \text{ cm}^3$$

The volume of the critical radius is given by

$$V_{r^*} = \frac{4}{3} \pi r^3 = \left(\frac{4}{3} \pi \right) (12.51 \times 10^{-8})^3 = 8200 \times 10^{-24} \text{ cm}^3$$

The number of unit cells in the critical nucleus is

$$\frac{V_{\text{unit cell}}}{V_{r^*}} = \frac{47.24 \times 10^{-24}}{8200 \times 10^{-24}} = 174 \text{ unit cells}$$

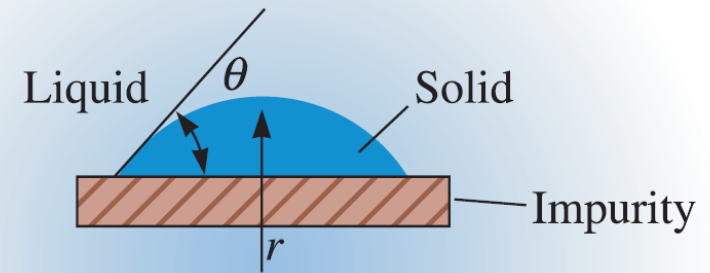
Since there are four atoms in each FCC unit cell, the number of atoms in the critical nucleus must be

$$(4 \text{ atoms/cell})(174 \text{ cells/nucleus}) = 696 \text{ atoms/nucleus}$$

In these types of calculations, we assume that a nucleus that is made from only a few hundred atoms still exhibits properties similar to those of bulk materials. This is not strictly correct and as such is a weakness of the classical theory of nucleation.

Heterogeneous Nucleation

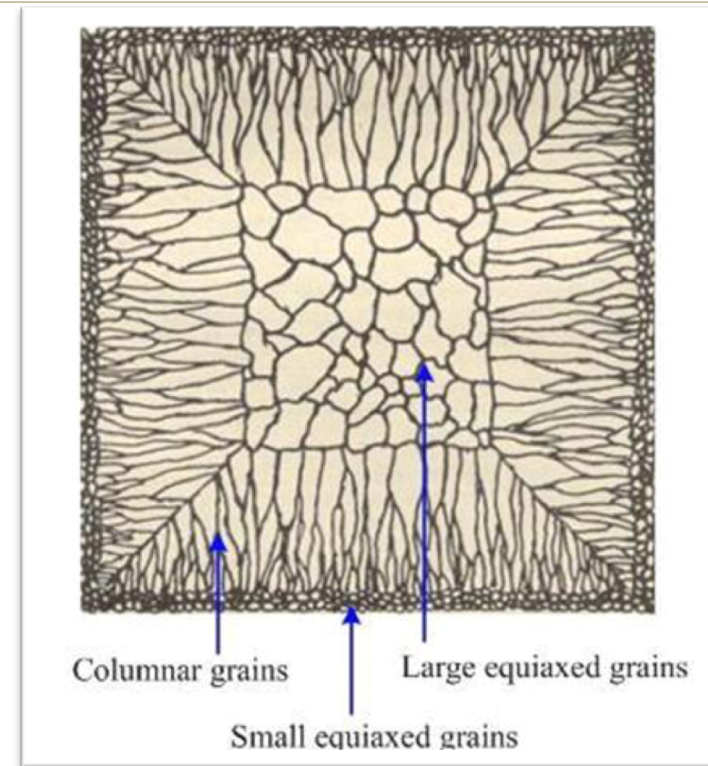
- A radius of curvature greater than the critical radius is achieved with very little total surface between the solid and liquid. Relatively few atoms must cluster together to produce a solid particle that has the required radius of curvature. Much less undercooling is required to achieve the critical size, so nucleation occurs more readily.
- Nucleation on preexisting surfaces is known as **heterogeneous nucleation**.
- This process is dependent on the contact angle (θ) for the nucleating phase and the surface on which nucleation occurs



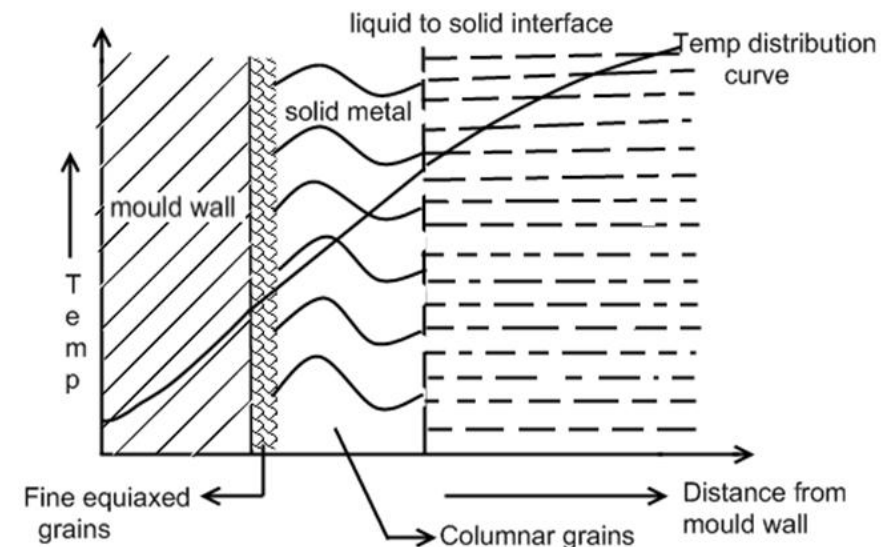
A solid forming on an impurity can be assumed to have the critical radius with a smaller increase in the surface energy. Thus, heterogeneous nucleation can occur with relatively low undercoolings.

Heterogeneous Nucleation

- Heterogeneous solidification occurs at mold walls and impurities.
- With heterogeneous solidification, the solid is not uniform, instead having three distinct regions within the solid.
- These regions are the chill zone, columnar zone, and equiaxed zone.
- The chill zone is nearest to the mold wall and contains a band of randomly oriented grains.
- The columnar zone consists of long columnar grains that grow opposite to the direction of heat flow.
- The equiaxed zone is in the center of the mold and has uniform equiaxed grains and controlled nucleation.



Grain Structure of Ingots



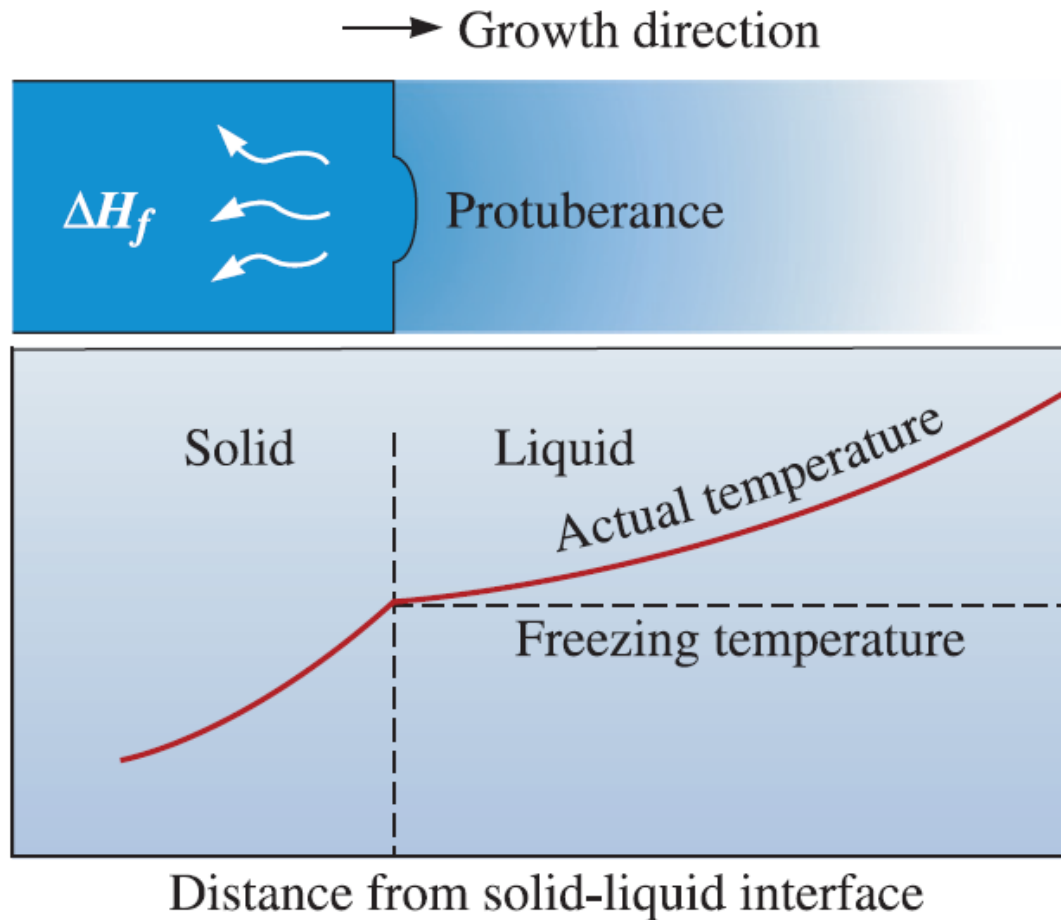
Applications of Controlled Nucleation

- ***Grain refinement*** - The addition of heterogeneous nuclei in a controlled manner to increase the number of grains in a casting.
- ***Dispersion strengthening*** - Increase in strength of a metallic material by generating resistance to dislocation motion by the introduction of small clusters of a second material.
- ***Solid-state phase transformation*** - A change in phase that occurs in the solid state.
- ***Rapid solidification processing*** - Producing unique material structures by promoting unusually high cooling rates during solidification.

Growth Mechanisms

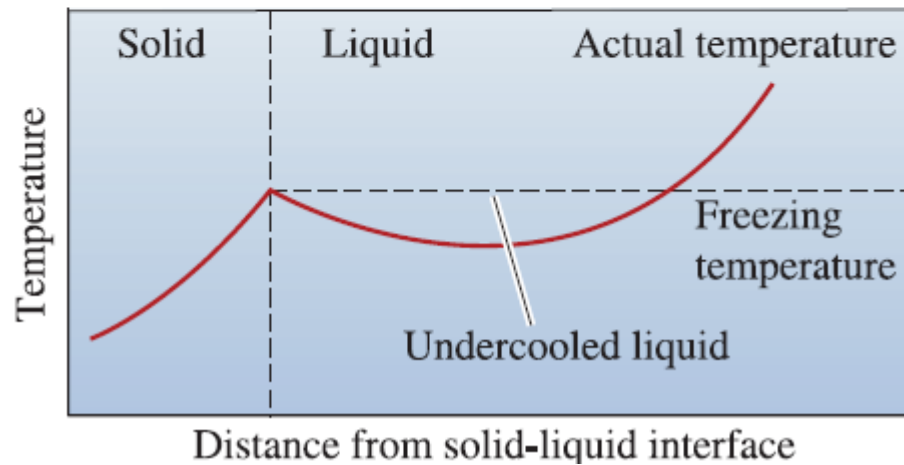
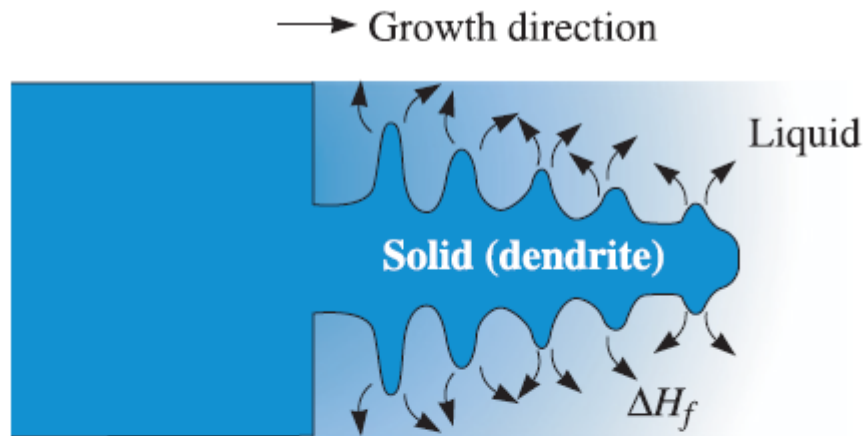
- Planar growth - The growth of a smooth solid-liquid interface during solidification, when no undercooling of the liquid is present.
- Dendrite - The treelike structure of the solid that grows when an undercooled liquid solidifies
- Specific heat - The heat required to change the temperature of a unit weight of the material one degree.
- Solidification front - Interface between a solid and liquid.

Growth Mechanisms - Planar growth



When the temperature of the liquid is above the freezing temperature, a protuberance on the solid-liquid interface will not grow, leading to maintenance of a planar interface. Latent heat is removed from the interface through the solid

Growth Mechanisms - Dendritic Growth

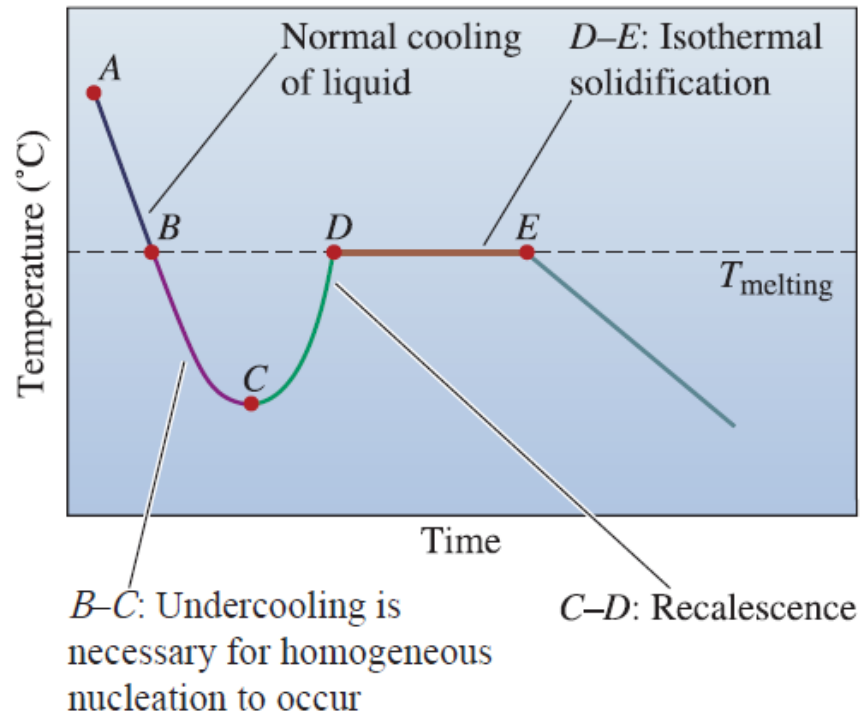


If the liquid is undercooled, a protuberance on the solid-liquid interface can grow rapidly as a dendrite. The latent heat of fusion is removed by raising the temperature of the liquid back to the freezing temperature.

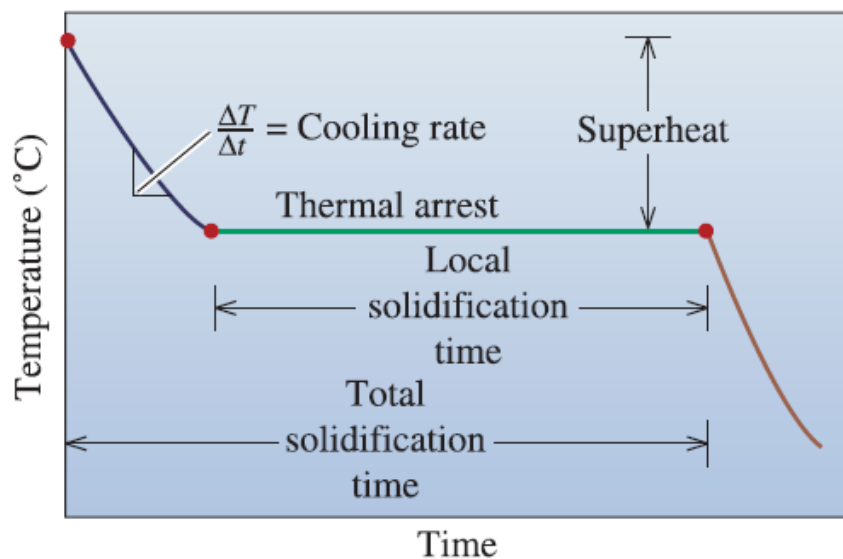


Scanning electron micrograph of dendrites in steel (x15).

Cooling Curves



Cooling curve for a pure metal that has not been well-inoculated. The liquid cools as specific heat is removed (between points A and B). Undercooling is thus necessary (between points B and C). As the nucleation begins (point C), latent heat of fusion is released causing an increase in the temperature of the liquid. This process is known as **recalescence** (point C to point D). The metal continues to solidify at a constant temperature (T_{melting}). At point E, solidification is complete. The solid casting continues to cool from this point.



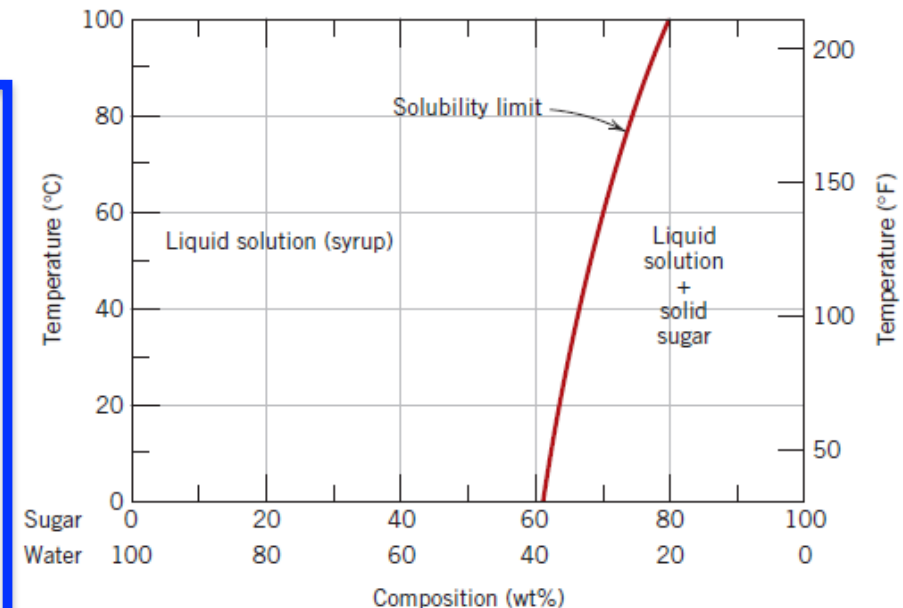
Cooling curve for a well-inoculated, but otherwise pure, metal. No undercooling is needed. Recalescence is not observed. Solidification begins at the melting temperature.

Definitions and Basic Concepts

Solubility Limit: At some specific temperature, there is a maximum concentration of solute atoms that may dissolve in the solvent to form a solid solution.

Phase: A region in a material that differs in structure and function from other regions.

- homogeneous in crystal structure and atomic arrangement
- have same chemical and physical properties throughout
- have a definite interface and able to be mechanically separated from its surroundings
- a single-phase system is termed homogeneous. Systems composed of two or more phases are termed mixtures or heterogeneous systems



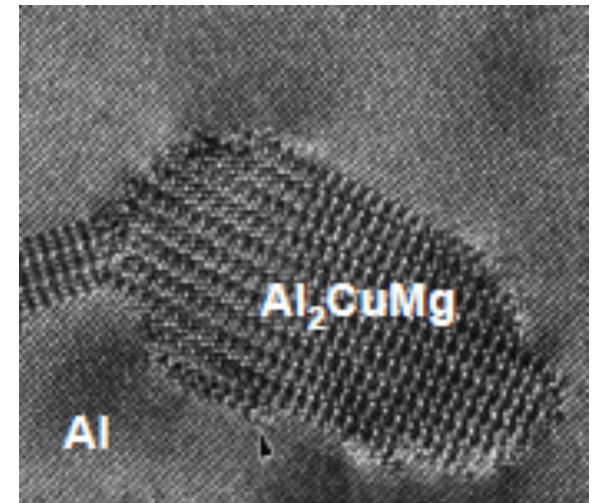
Definitions and Basic Concepts

Phase diagrams or equilibrium diagram is a type of graph used to show the equilibrium conditions between the thermodynamically-distinct phases; or to show what phases are present in the material system at various **temperature, pressure, and compositions**

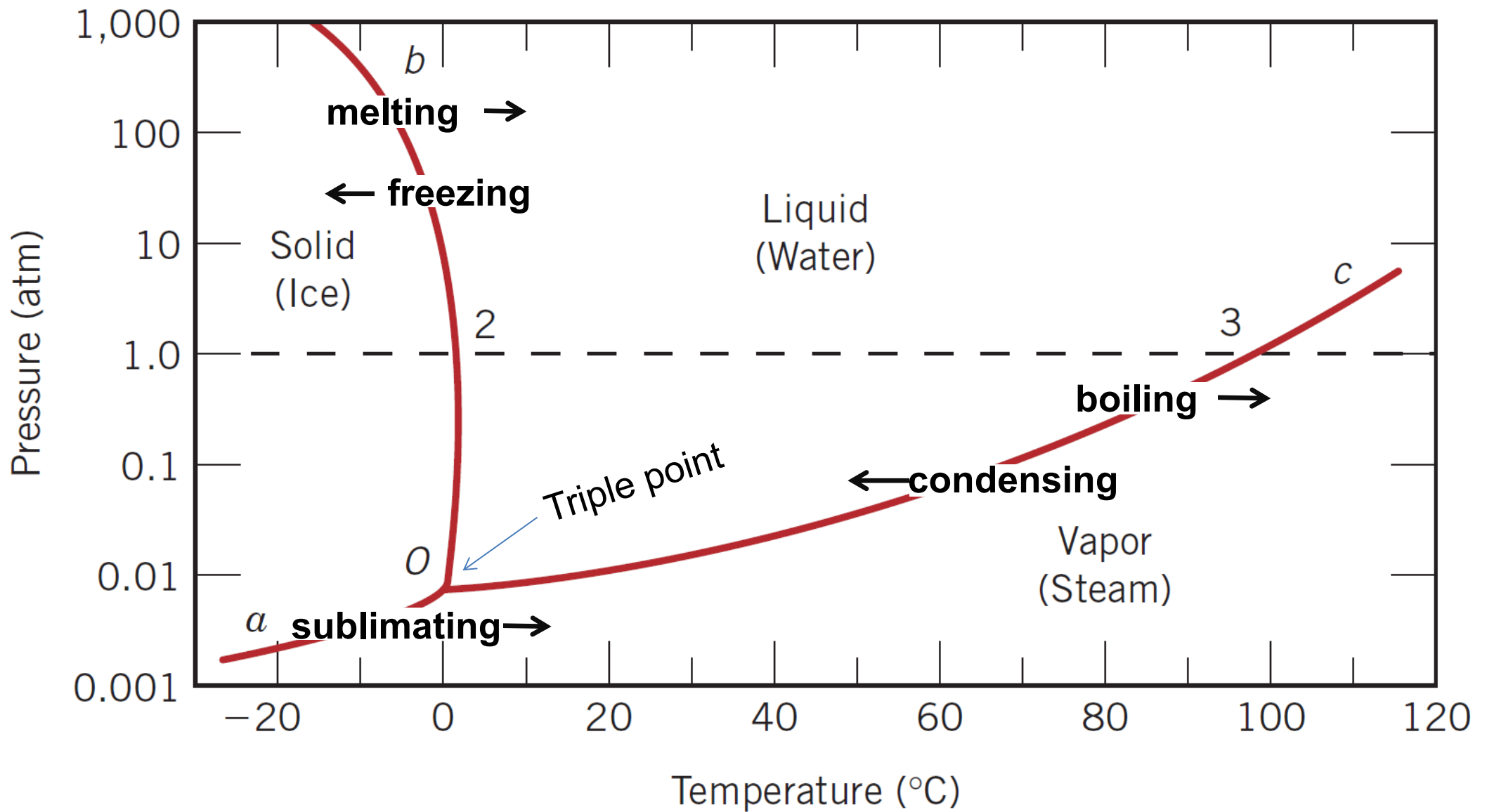
- Indicates equilibrium solid solubility of one element in another.
- Indicates temperature range under which solidification occurs.
- Indicates temperature at which different phases start to melt

Note: •“equilibrium” is important: phase diagrams are determined by using slow cooling conditions $\Rightarrow$ no information about kinetics

Degree of freedom (or variance) F is the number of variables (temperature, pressure, and composition) that can be changed independently without changing the phases of the system



Phase Diagram of Water



aO, *bO*, and *cO* are phase boundaries

Gibbs Phase Rule

- From thermodynamic consideration, J. W. Gibbs derived an equation that computes the number of phases that can coexist in equilibrium in a chosen system. This equation, called Gibbs phase rule, is

$$P + F = C + 2$$

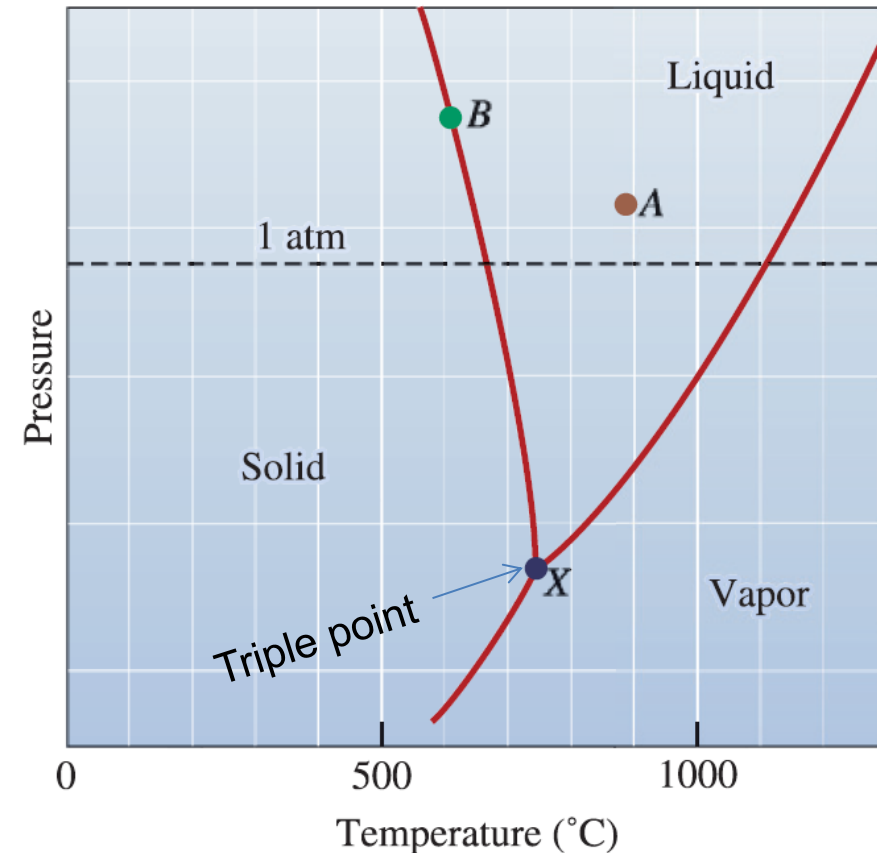
Where P is number of phases that coexist in a system,
C = Number of components and
F = Degrees of freedom

For pure water, at triple point, 3 phases coexist.

- There is one component (water) in the system.
- Therefore $3 + F = 1 + 2$ $F = 0$.
- Degrees of freedom indicate number of variables that can be changed without changing number of phases.

Unary phase diagram (one component, $C = 1$)

- consider the case of pure magnesium (Mg).
- Depending on the temperature and pressure, however, there may be one, two, or even three phases present at any one time: solid magnesium, liquid magnesium, and magnesium vapor.
- At one atmosphere, (dashed line), the intersection in the phase diagram give the usual melting and boiling temperatures for magnesium.
- At very low pressures, a solid such as magnesium (Mg) can sublime, or go directly to a vapor form without melting, when it is heated.
- Suppose we have a pressure and temperature that put us at point A in the phase diagram (Figure). At this point, magnesium is all liquid. The number of phases is one (liquid). The phase rule tells us that there are two degrees of freedom



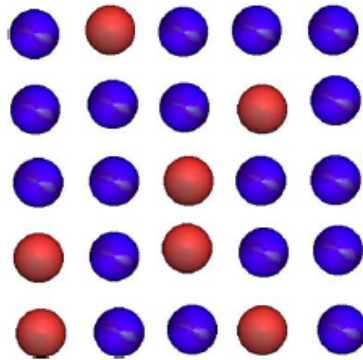
$$2 + C = F + P, \text{ therefore, } 2 + 1 = F + 1 \text{ (i.e., } F = 2\text{)}$$

Solubility and Solid Solutions

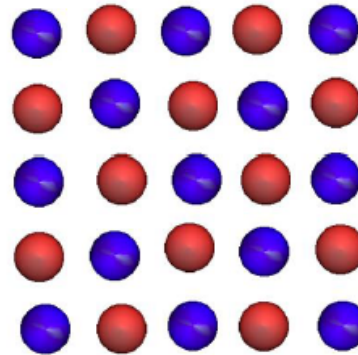
- When two metals are mixed together they form an alloy if one metal is soluble in the other one in solid state. Therefore, an alloy is a solid solution of two or more metals.
- Primarily there are two types of solid solutions -

Substitutional – Solute atoms occupy the regular lattice sites of the parent metal (solvent). Substitutional solid solutions can be random (Cu-Ni) or ordered (Cu-Au).

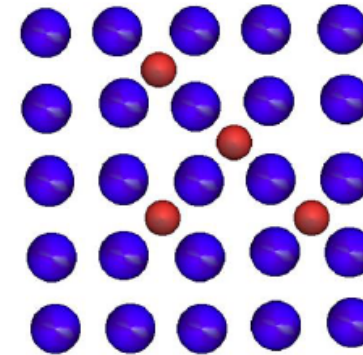
Interstitial – Solute atoms occupy the interstitial positions (Steel – C solute atoms in Fe) .



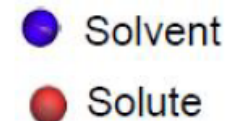
Random substitutional



Ordered substitutional



Interstitial

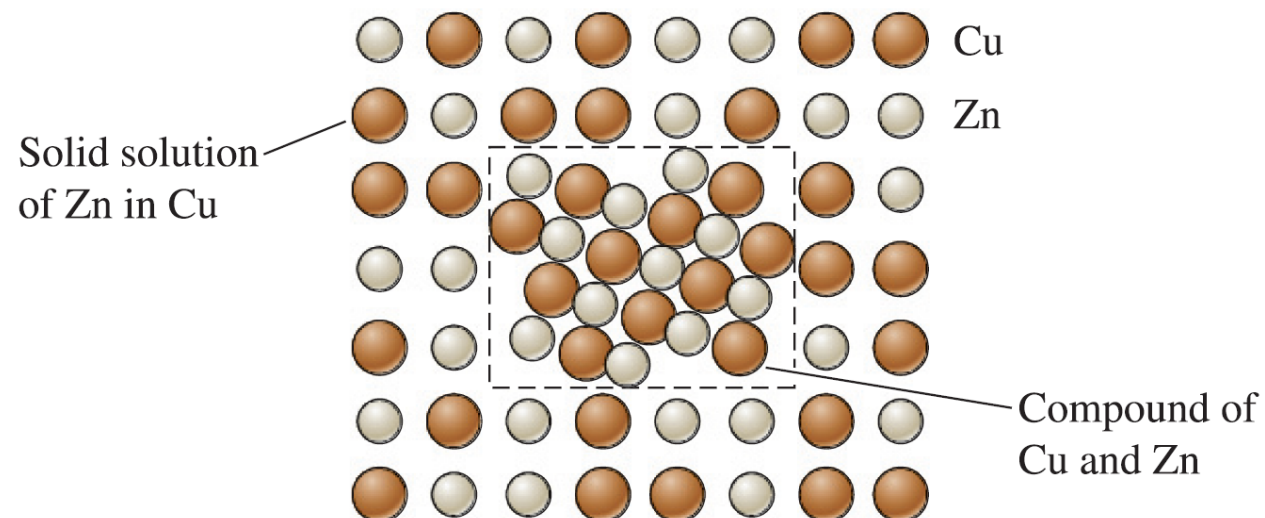


Solubility and Solid Solutions - *Limited Solubility*

Limited Solubility: If we add a small amount of liquid zinc to liquid copper, a single liquid solution is produced. When that copper-zinc solution cools and solidifies, a single solid solution having an FCC structure results, with copper and zinc atoms randomly located at the normal lattice points.

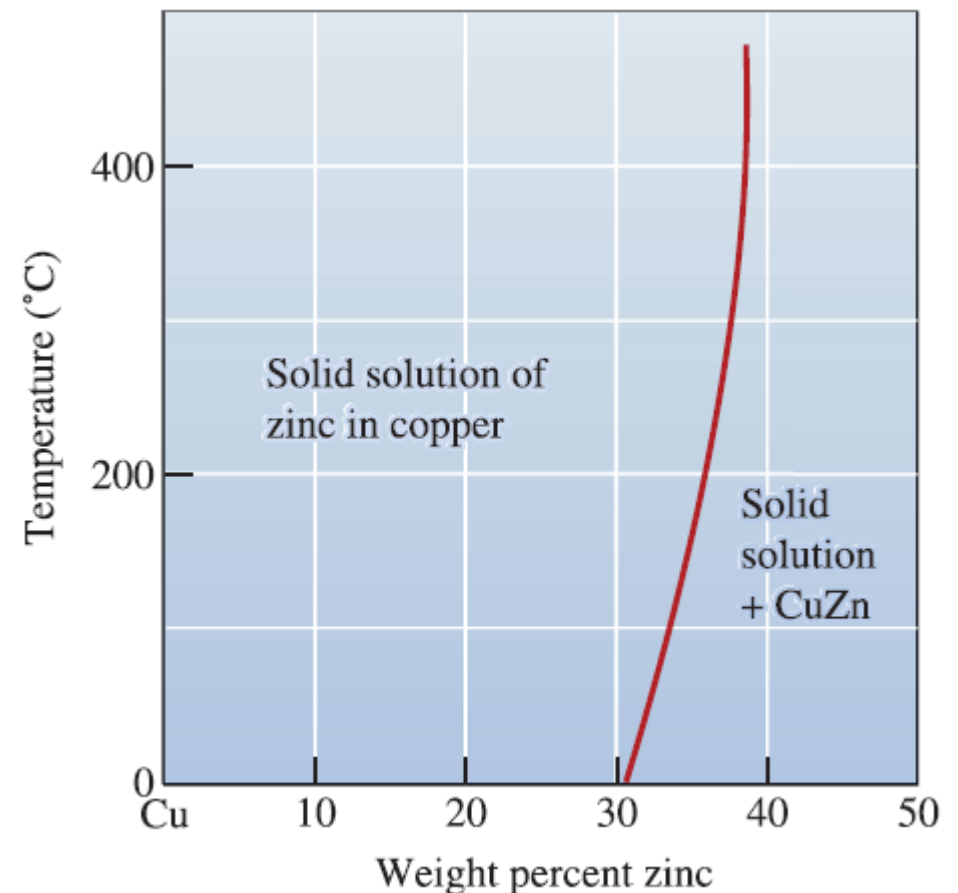
- If the liquid solution contains more than about 30% Zn, some of the excess zinc atoms combine with some of the copper atoms to form a CuZn compound [Figure].

- Two solid phases now coexist: a solid solution of copper saturated with about 30% Zn plus a CuZn compound.



Solubility and Solid Solutions - *Limited Solubility*

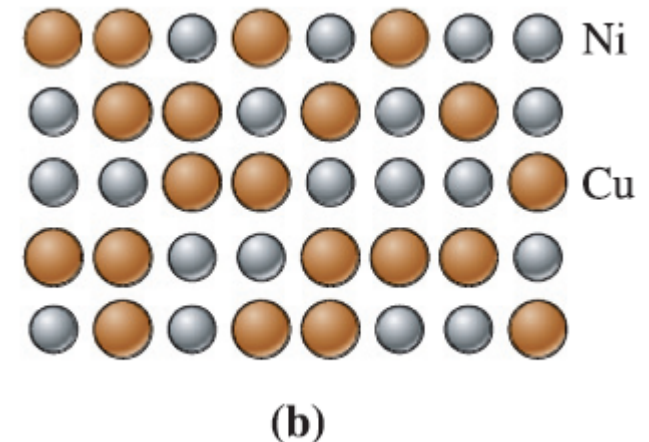
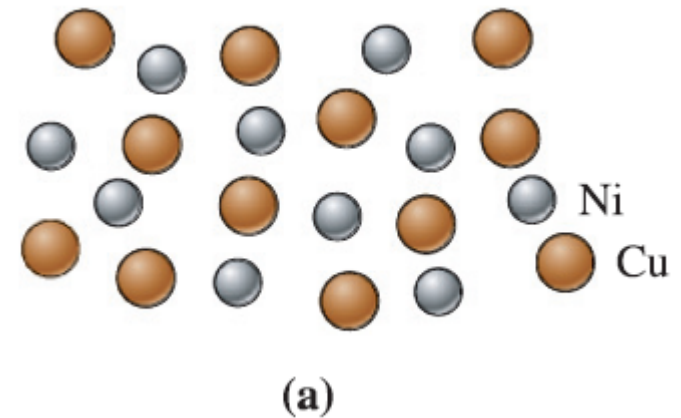
- The solubility of zinc in copper is limited.
- Figure shows a portion of the Cu-Zn phase diagram illustrating the solubility of zinc in copper at low temperatures. The solubility increases with increasing temperature



The solubility of zinc in copper. The solid line represents the solubility limit; when excess zinc is added, the solubility limit is exceeded and two phases coexist.

Solubility and Solid Solutions - *Unlimited Solubility*

- Now consider liquid copper and liquid nickel, only one liquid phase would be produced, when we mix any amounts.
- This liquid alloy has the same composition and properties everywhere [Figure (a)] because nickel and copper have ***unlimited liquid solubility***.
- If the liquid copper-nickel alloy solidifies and cools to room temperature while maintaining thermal equilibrium, only one solid phase is produced. After solidification, the copper and nickel atoms do not separate but, instead, are randomly located within the FCC crystal structure. Within the solid phase, the structure, properties, and composition are uniform and no interface exists between the copper and nickel atoms. Therefore, copper and nickel also have unlimited solid solubility. The solid phase is a solid solution of copper and nickel. [Figure (b)]



Solubility and Solid Solutions - *Unlimited Solubility*

- A solid solution is not a mixture.
- A mixture contains more than one type of phase, and the characteristics of each phase are retained when the mixture is formed. In contrast to this, the components of a solid solution completely dissolve in one another and do not retain their individual characteristics.

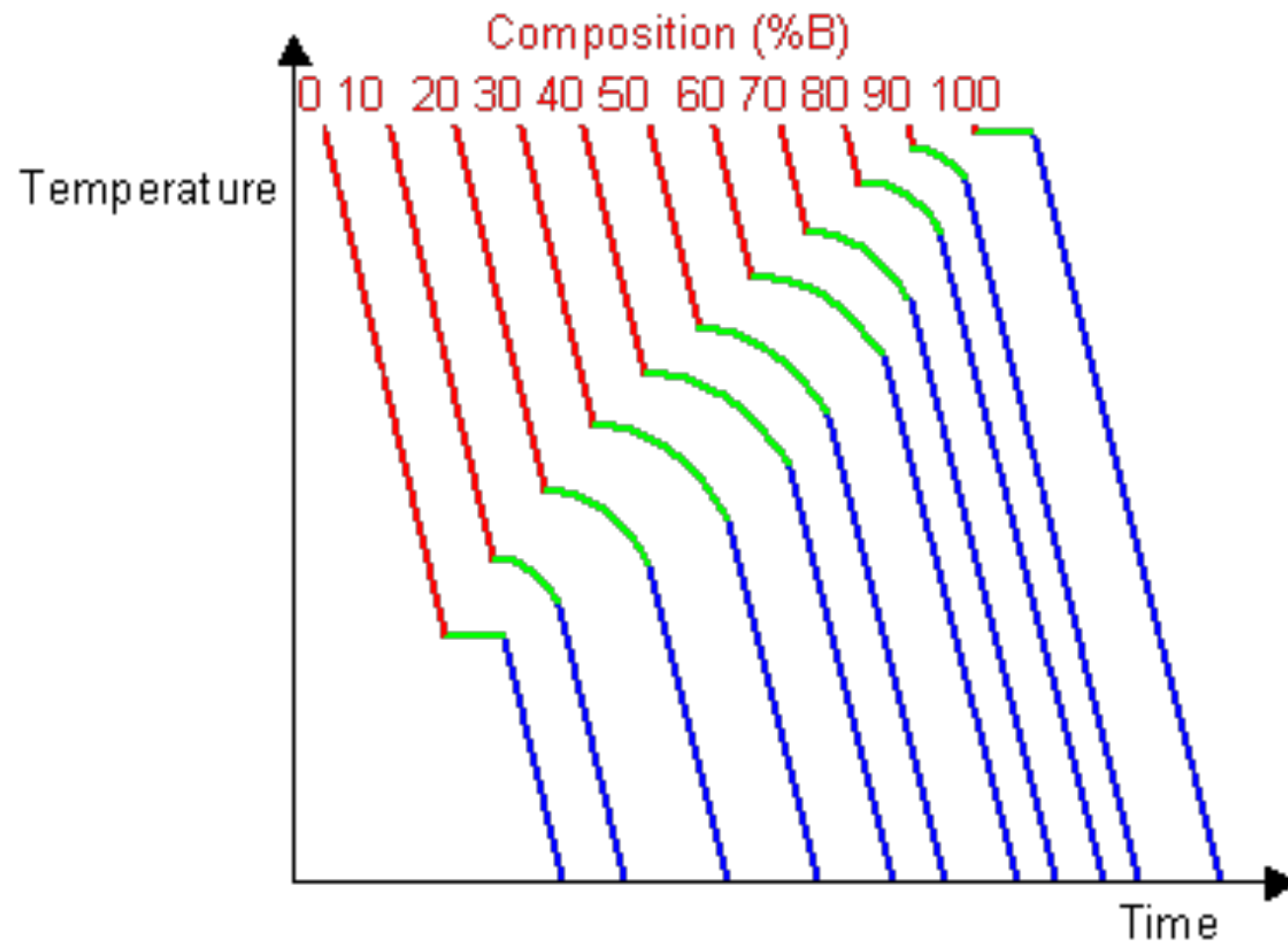
Conditions for Unlimited Solid Solubility

- In order for an alloy system, such as copper-nickel to have unlimited solid solubility, certain conditions must be satisfied. These conditions, the *Hume-Rothery rules*.

Hume-Rothery rules

The Hume-Rothery rules, are as follows:

1. **Size factor:** The atoms or ions must be of similar size, with no more than a 15% difference in atomic radius, in order to minimize the lattice strain (i.e., to minimize, at an atomic level, the deviations caused in interatomic spacing).
2. **Crystal structure:** The materials must have the same crystal structure; otherwise, there is some point at which a transition occurs from one phase to a second phase with a different structure.
3. **Valence:** The ions must have the same valence; otherwise, the valence electron difference encourages the formation of compounds rather than solutions.
4. **Electronegativity:** The atoms must have approximately the same electronegativity. Electronegativity is the affinity for electrons. If the electronegativities differ significantly, compounds form—as when sodium and chloride ions combine to form sodium chloride.



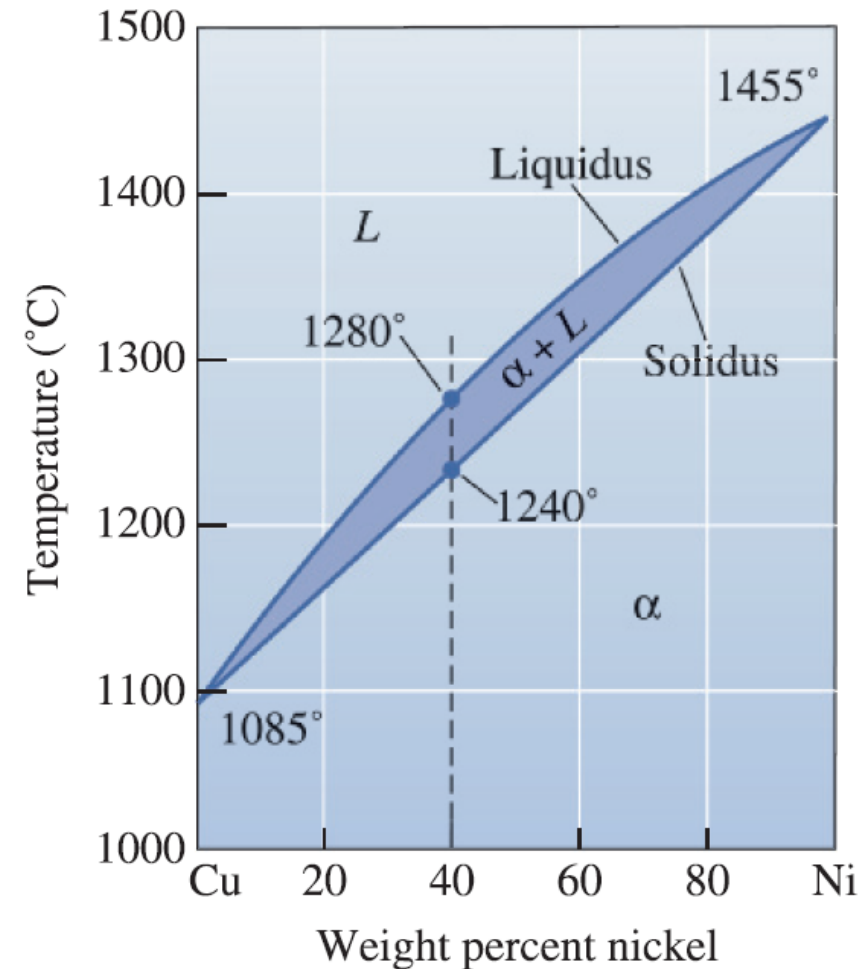
Binary Isomorphous alloy system

- A phase diagram shows the phases and their compositions at any combination of temperature and alloy composition.
- When only two elements or two compounds are present in a material, a binary phase diagram can be constructed.
- Isomorphous phase diagrams are found in a number of metallic and ceramic systems.
- Possibly the easiest type of binary phase diagram to understand and interpret is the type that is characterized by the copper–nickel system.
- The copper–nickel system is termed ***isomorphous*** because of this complete liquid and solid solubility of the two components.

Binary Isomorphous alloy system

Nomenclature

- First, for metallic alloys, solid solutions are commonly designated by lowercase Greek letters (α , β , γ etc.).
- Furthermore, with regard to phase boundaries, the line separating the L and $\alpha + L$ phase fields is termed the **liquidus line**.
- The liquid phase is present at all temperatures and compositions above this line.
- The **solidus line** is located between the α and $\alpha + L$ regions, below which only the solid phase exists.



Binary Isomorphous alloy system

Interpretation of phase diagrams

For a binary system of known composition and temperature that is at equilibrium, at least three kinds of information are available:

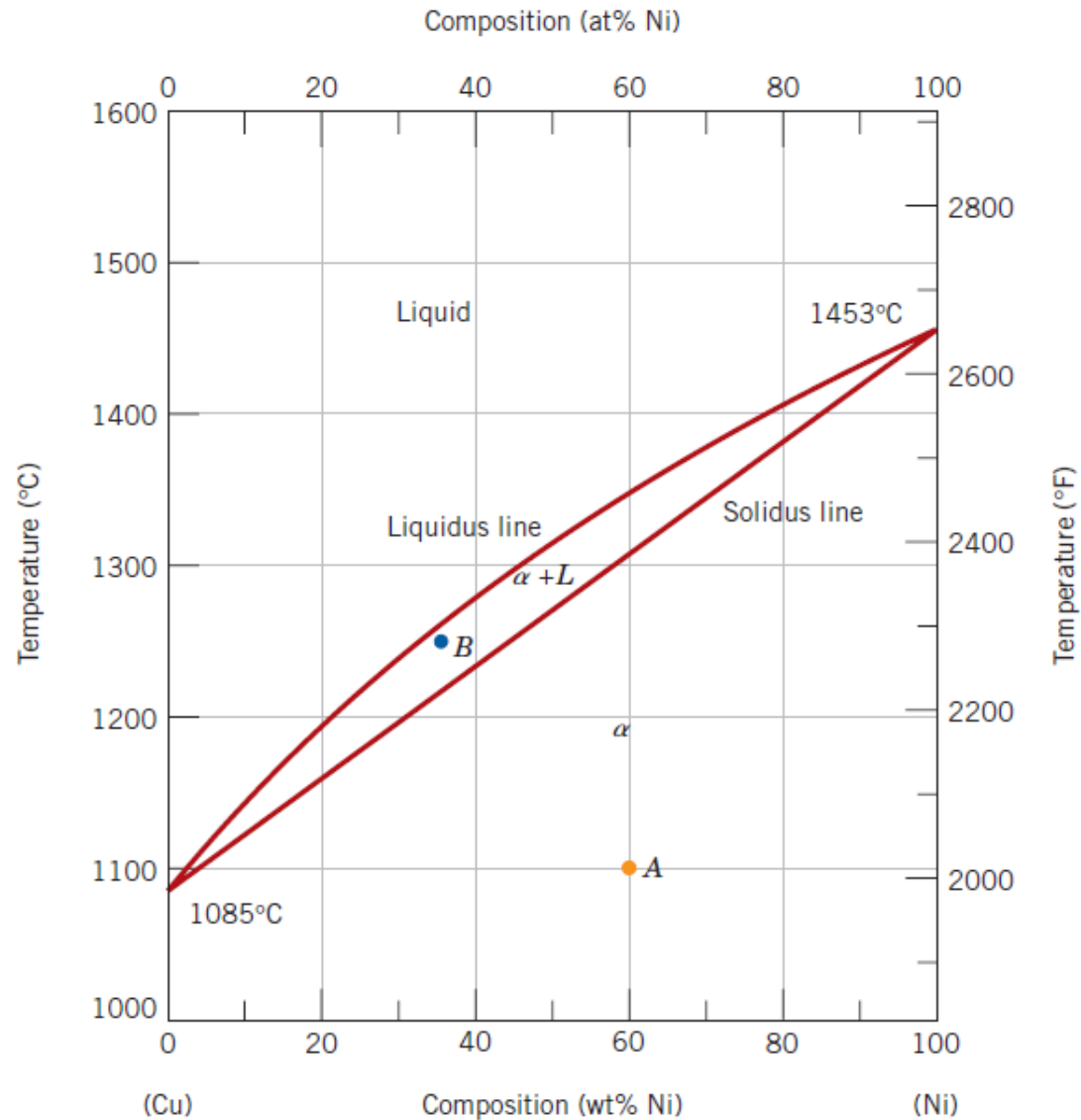
1. the phases that are present,
2. the compositions of these phases, and
3. the percentages or fractions of the phases.

The procedures for making these determinations will be demonstrated using the copper-nickel system.

Interpretation of phase diagrams

Phases Present: *If we know or locate the temperature–composition point, then we know the number and types of the phase(s) present.*

- For example, an alloy of composition 60 wt% Ni–40 wt% Cu at 1100 °C would be located at point A in Figure; because this is within the α region, only the single α phase will be present.
- On the other hand, a 35 wt% Ni–65 wt% Cu alloy at 1250°C (point B) will consist of both α and liquid phases at equilibrium.



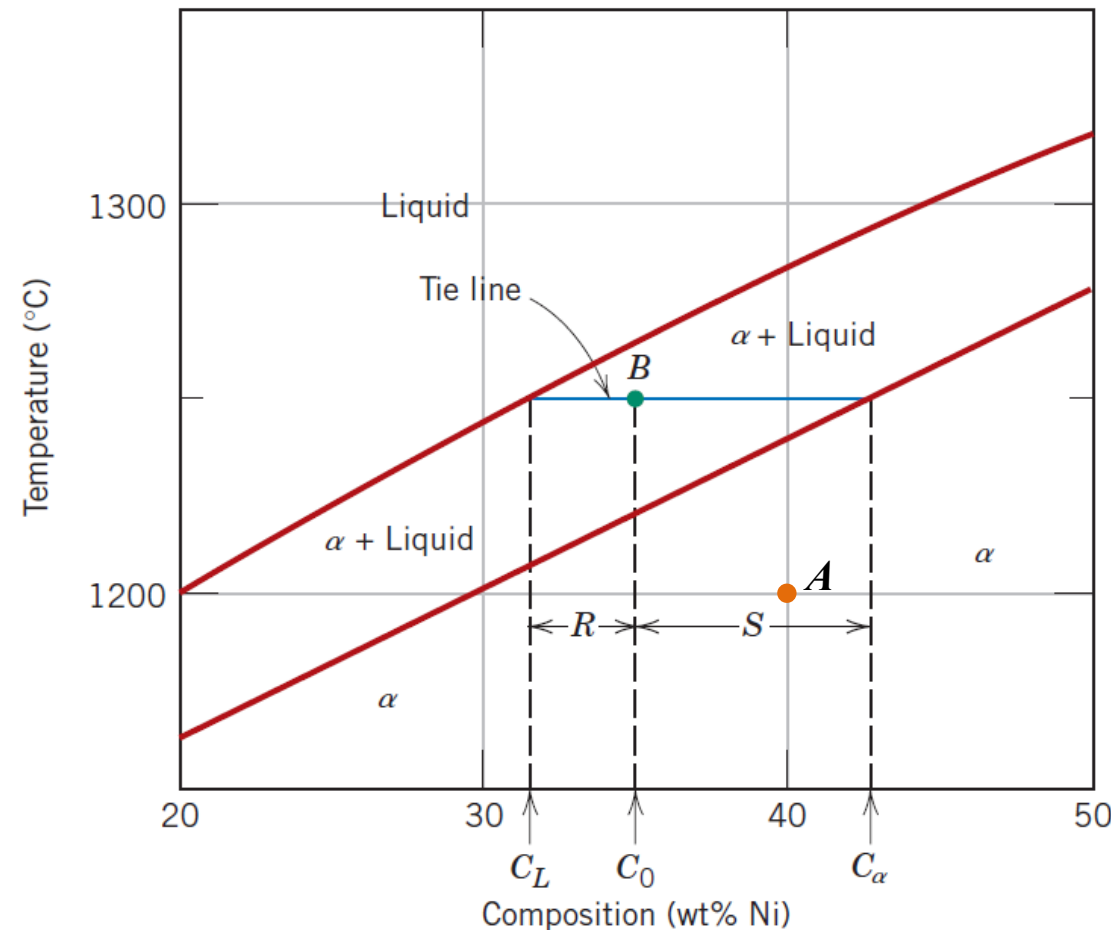
Interpretation of phase diagrams

Determination of Phase Compositions: *If we know or locate the temperature–composition point, then we know the composition of each phase present.*

Different methods are used for determination of single- and two-phase regions

Single- phase regions

For example, consider the 40 wt% Ni–60 wt% Cu alloy at 1100°C (point A, Figure). At this composition and temperature, only the phase is present, having a composition of 40 wt% Ni–60 wt% Cu.

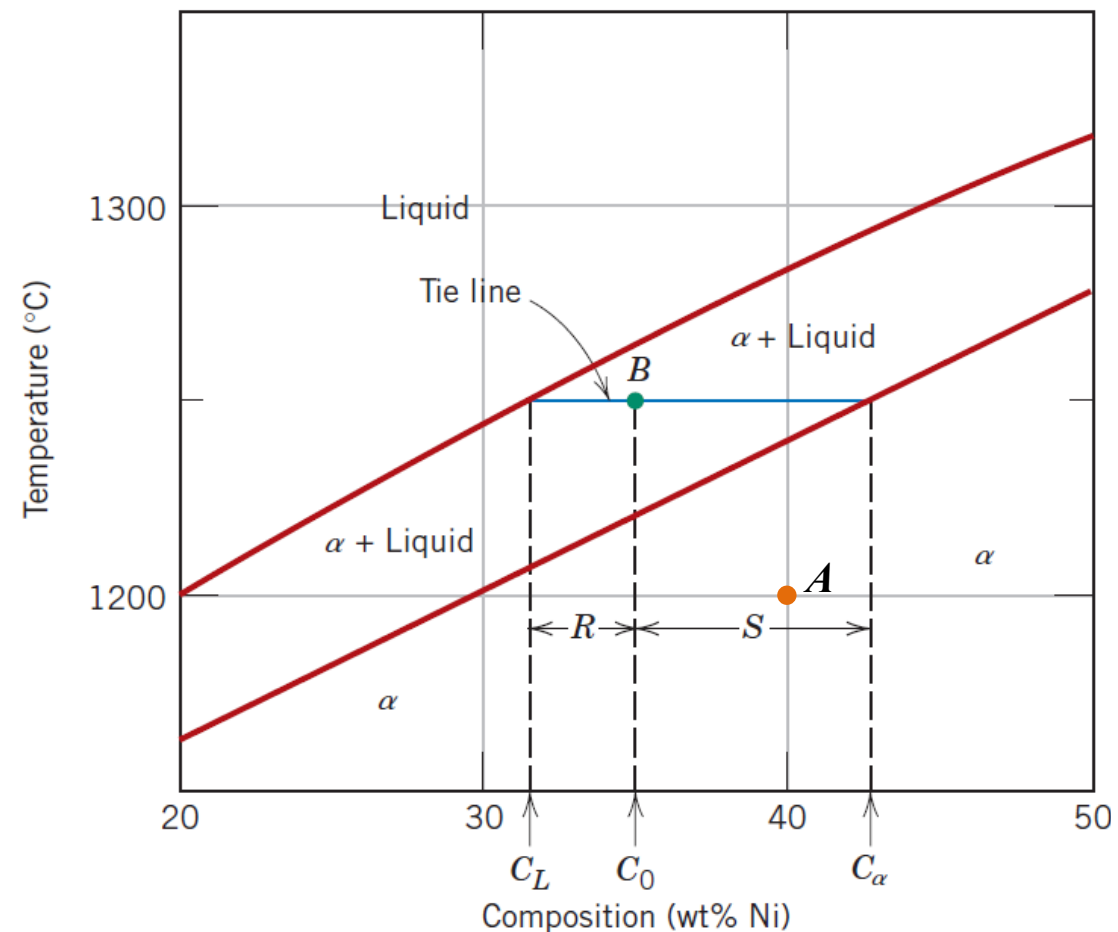


Interpretation of phase diagrams

Determination of Phase Compositions

Two-phase regions

- imagine a horizontal lines (**tie line**, or sometimes as an **isotherm**) passing through the temperature–composition point and terminate at the phase boundary lines on either side.
- To compute the equilibrium concentrations of the two phases, the following procedure is used



Interpretation of phase diagrams

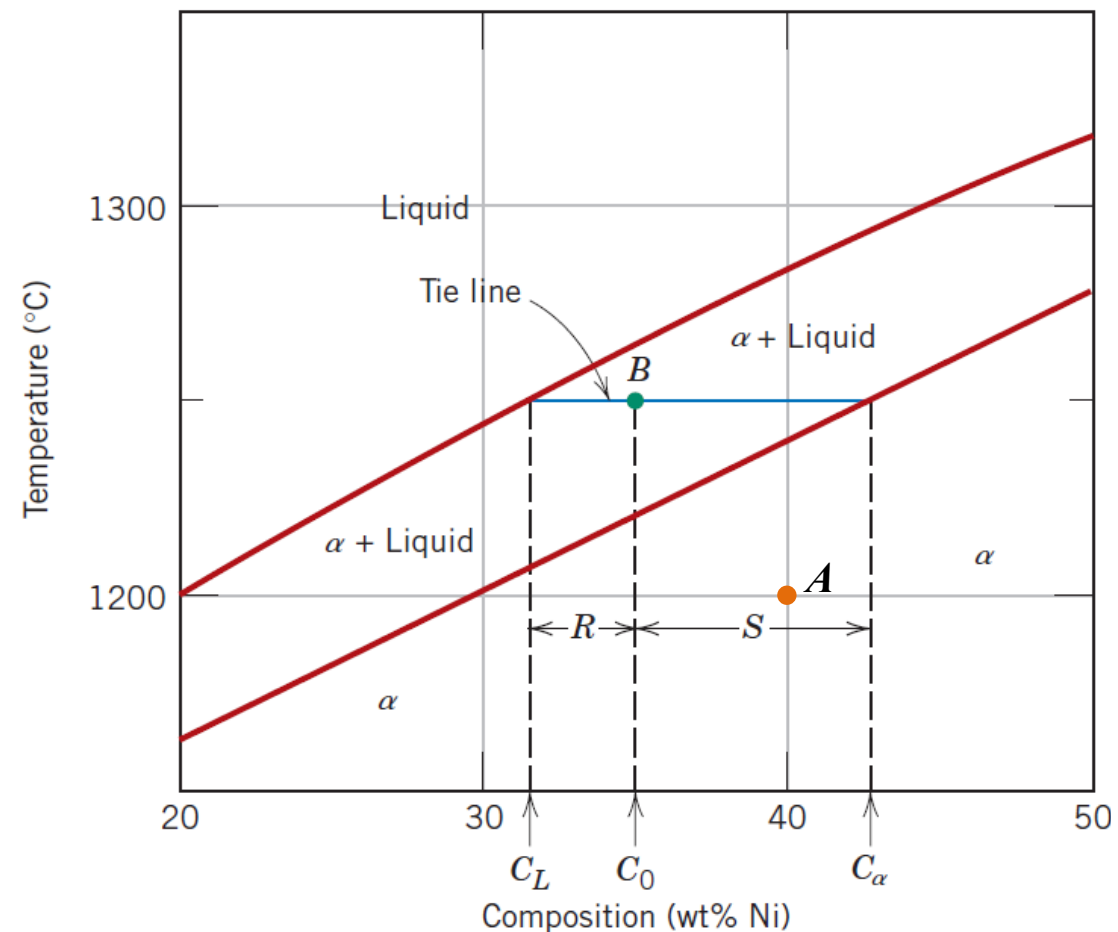
Determination of Phase Compositions

1. A tie line is constructed across the two-phase region at the temperature of the alloy.
2. The intersections of the tie line and the phase boundaries on either side are noted.
3. Perpendiculars are dropped from these intersections to the horizontal composition axis, from which the composition of each of the respective phases is read.

In figure, at point B, after tie line intersection,

$$C_L = C_{\text{liquidus}} = 31.5 \text{ wt\% Ni} - 68.5 \text{ wt\% Cu}$$

$$C_\alpha = C_{\text{solidus}} = 42.5 \text{ wt\% Ni} - 57.5 \text{ wt\% Cu}$$



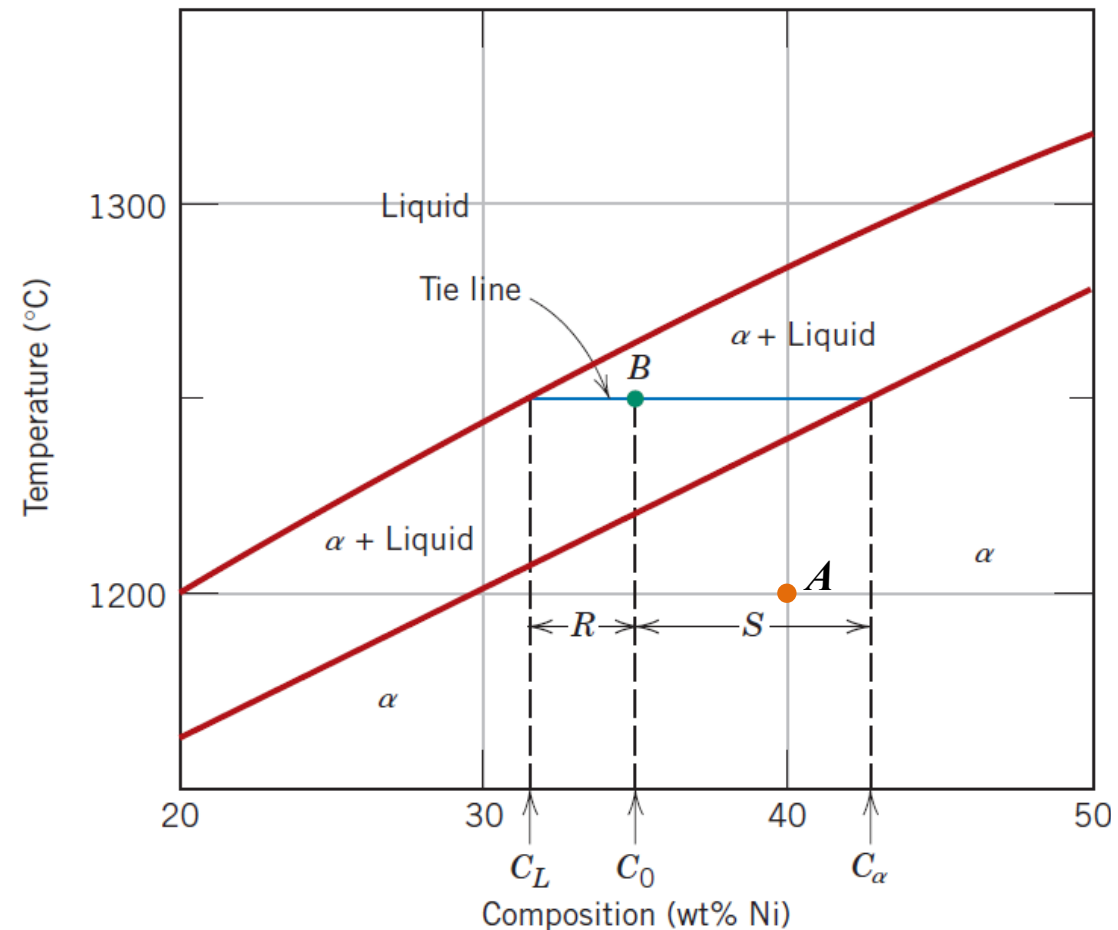
Interpretation of phase diagrams

Determination of Phase Amounts

For single-phase region:

because only one phase is present, the alloy is composed entirely of that phase; i.e., the phase fraction is 1.0 or, alternatively, the percentage is 100%.

For the 40 wt% Ni–60 wt% Cu alloy at 1100°C (point A in Figure), only the α phase is present; hence, the alloy is completely or 100% α



Interpretation of phase diagrams

Determination of Phase Amounts - For two-phase region,

The tie line must be utilized in conjunction with a procedure that is often called **the lever rule (or the inverse lever rule)**, which is applied as follows:

1. The tie line is constructed across the two-phase region at the temperature of the alloy.
2. The overall alloy composition is located on the tie line.
3. The fraction of one phase is computed by taking the length of tie line from the overall alloy composition to the phase boundary for the *other phase*, and dividing by the total tie line length.
4. The fraction of the other phase is determined in the same manner.
5. If phase percentages are desired, each phase fraction is multiplied by 100. When the composition axis is scaled in weight percent, the phase fractions computed using the lever rule are mass fractions—the mass (or weight) of a specific phase divided by the total alloy mass (or weight). The mass of each phase is computed from the product of each phase fraction and the total alloy mass.

Interpretation of phase diagrams

At 1250°C both α and liquid phases are present for a 35 wt% Ni–65 wt% Cu alloy.

- Let the overall alloy composition be located along the tie line and denoted as C_0 , and mass fractions be represented by W_L and W_α for the respective phases.

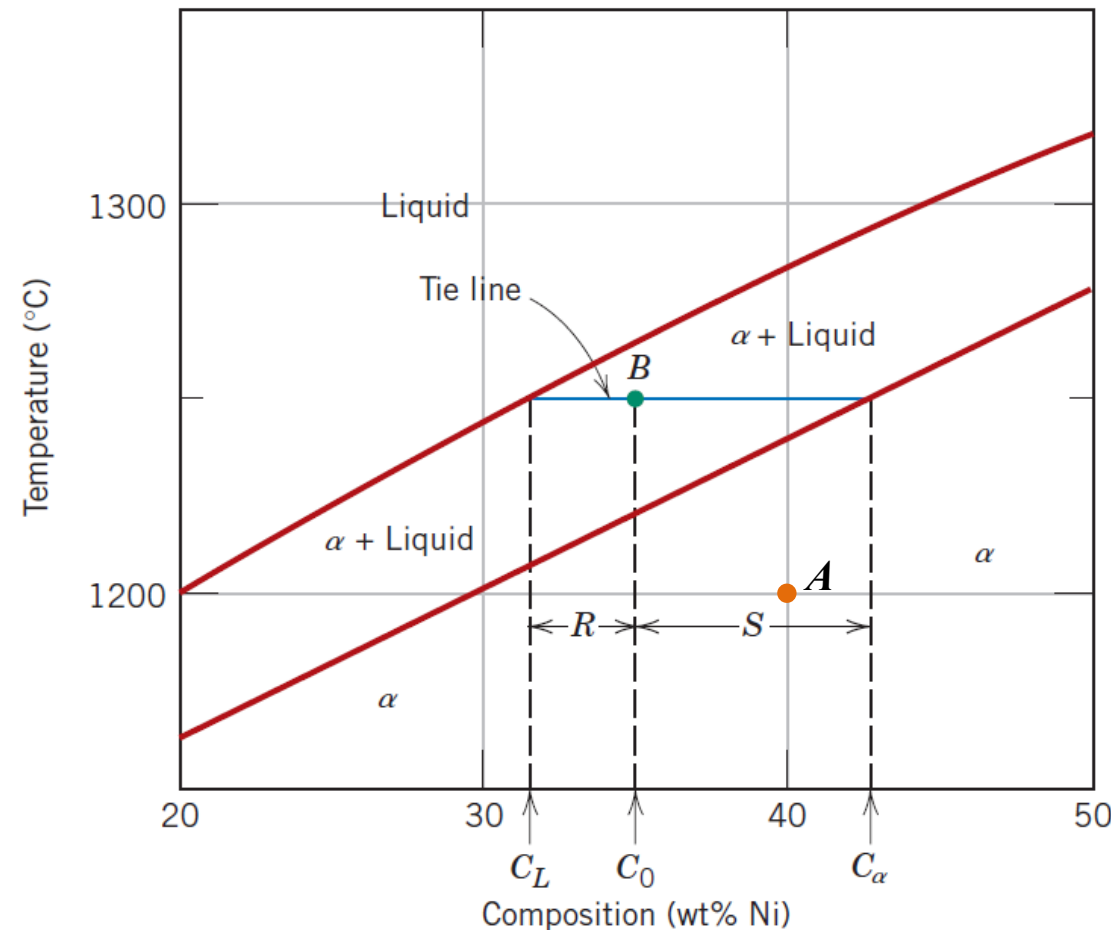
From the lever rule,

W_L may be computed according to

$$W_L = \frac{S}{R+S} = \frac{C_\alpha - C_0}{C_\alpha - C_L} = \frac{42.5 - 35}{42.5 - 31.5} = 0.68$$

W_α may be computed according to

$$W_\alpha = \frac{R}{R+S} = \frac{C_0 - C_L}{C_\alpha - C_L} = \frac{35 - 31.5}{42.5 - 31.5} = 0.32$$

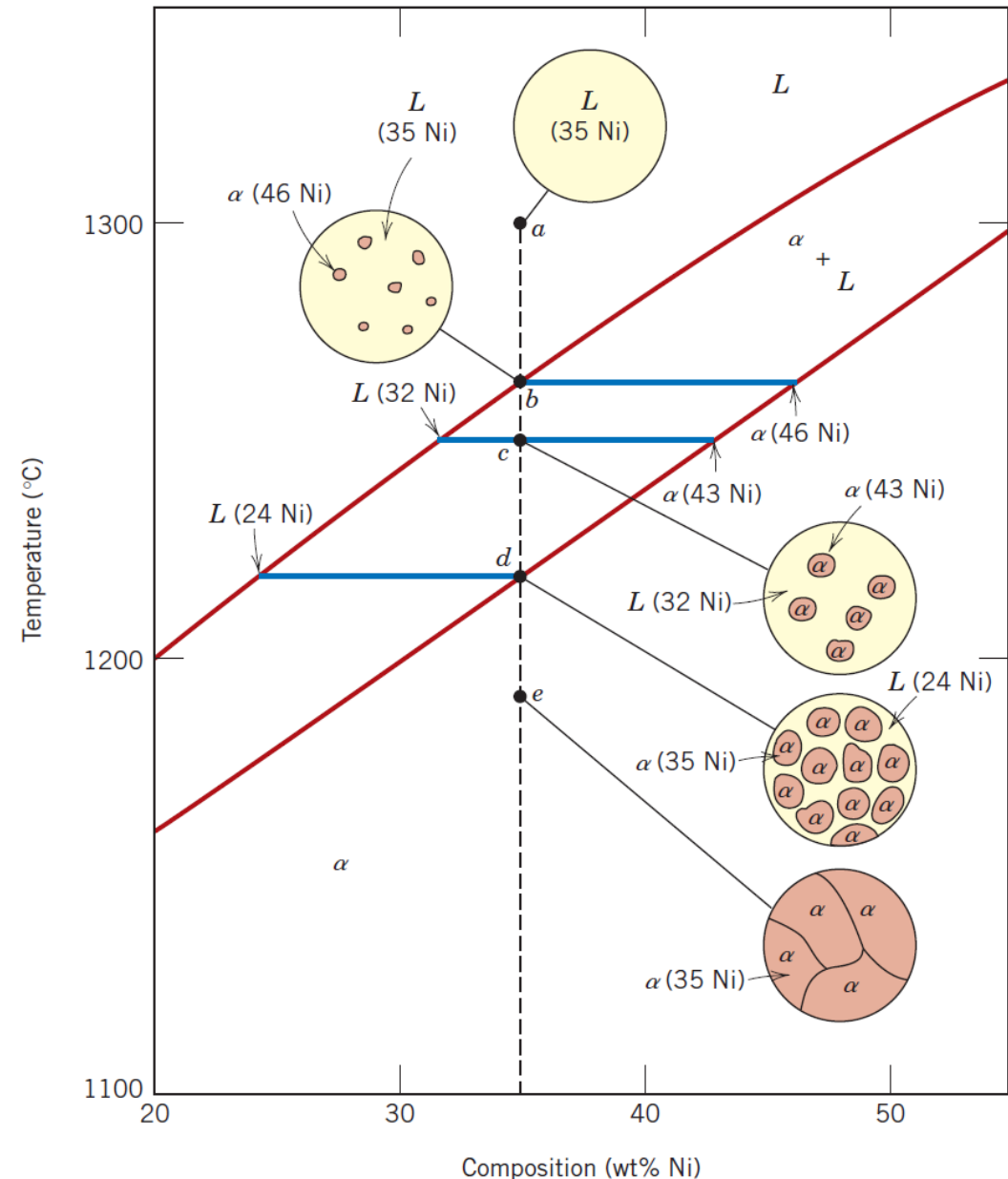


Isomorphous alloy system

Development of Microstructure

Equilibrium Cooling

- cooling occurs very slowly, in that phase equilibrium is continuously maintained
- At 1300°C, point *a*,
 - Liquid (35 wt% Ni–65 wt% Cu)
- At 1260°C, point *b*,
 - Liquid + α [46 wt% Ni–54 wt% Cu, α (46 Ni)]
- At 1250°C, point *c*,
 - Liquid [32 wt% Ni–68 wt% Cu [L(32 Ni)] + α and 43 wt% Ni–57 wt% Cu [α (43 Ni)],
- At 1220°C, point *d*,
 - Liquid 24 wt% Ni–76 wt% Cu [L(24 Ni)], + α [35 wt% Ni–65 wt% Cu [α (35Ni)]]
- At point *e*,
 - α [35 wt% Ni–65 wt% Cu [α (35Ni)]]

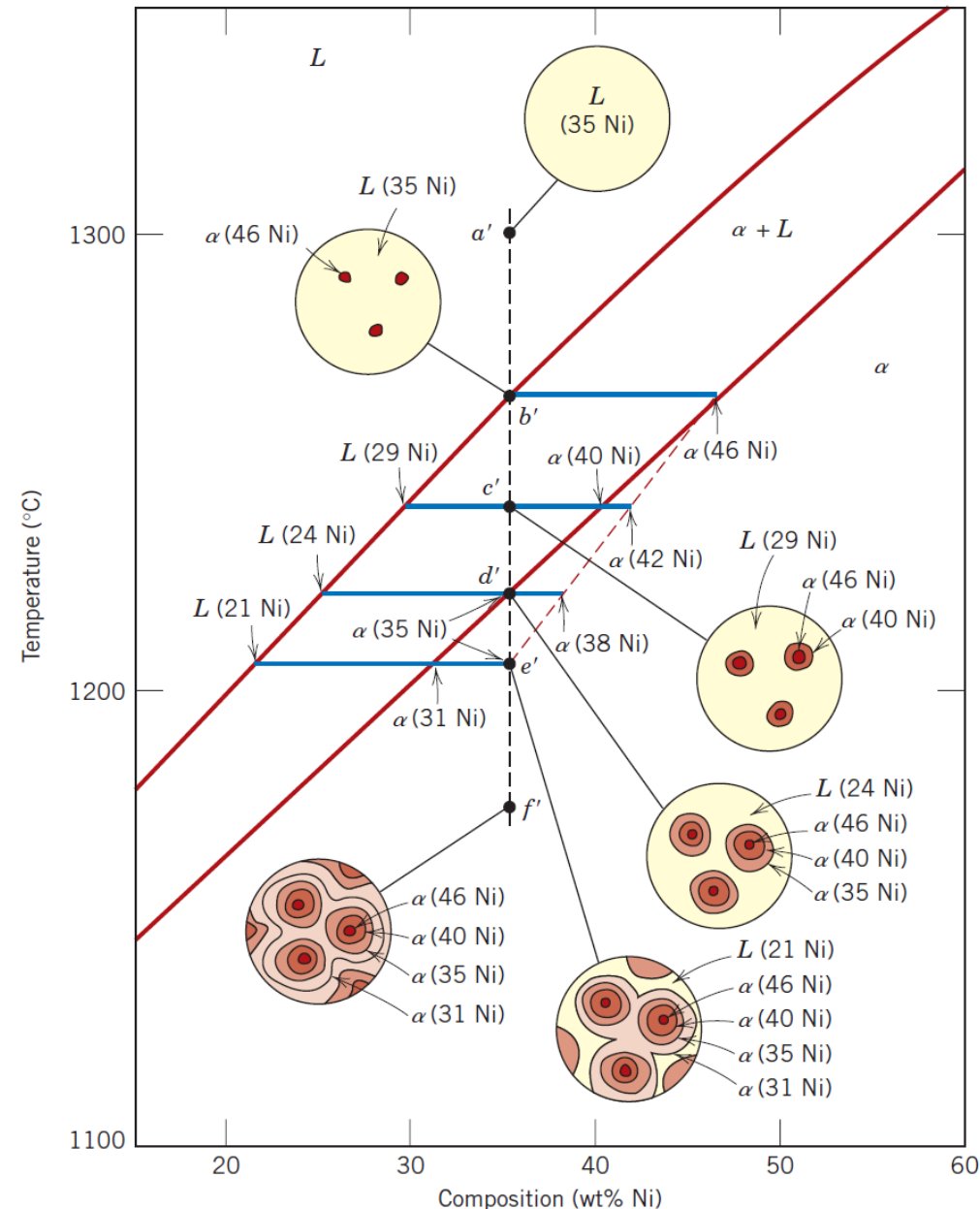


Isomorphous alloy system

Development of Microstructure

Nonequilibrium Cooling

- In virtually all practical solidification situations, cooling rates are much too rapid to allow these compositional readjustments and maintenance of equilibrium.
- Diffusion rates are especially low for the solid phase and, for both phases, decrease with diminishing temperature; consequently, microstructures other than those previously described develop.

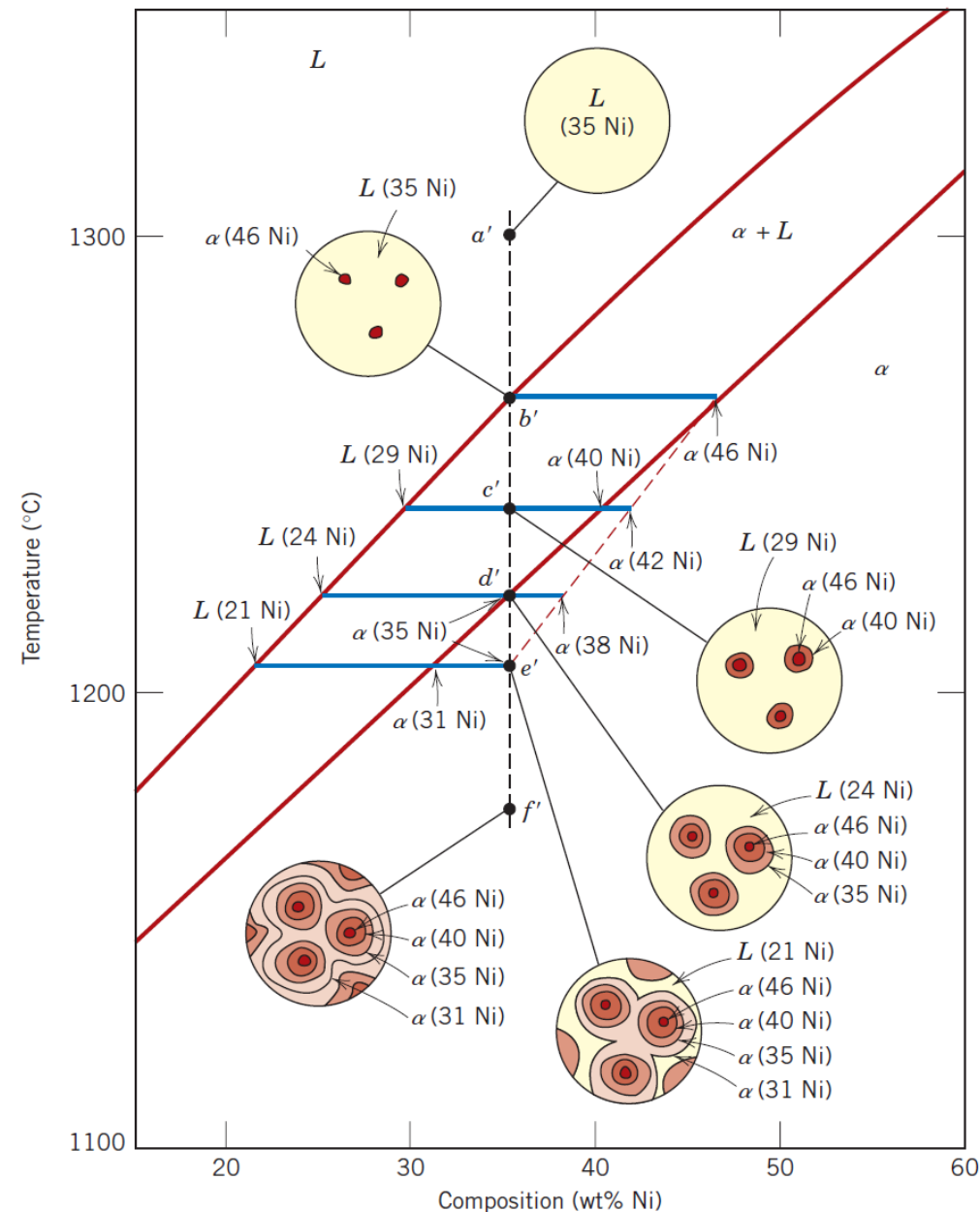


Isomorphous alloy system

Development of Microstructure

Nonequilibrium Cooling

- At 1300°C, point a',
➤ Liquid (35 wt% Ni–65 wt% Cu)
- At 1260°C, point b',
➤ Liquid + α [46 wt% Ni–54 wt% Cu, α (46 Ni)]
- At 1240°C, point c', **due to relatively slow diffusion in the solid α phase, nonequilibrium solidus temperature is followed.**
➤ L(29 Ni) + α (46 Ni) + α (40 Ni)
- Similarly at 1220°C, point d',
➤ L(24 Ni) + α (46 Ni) + α (40 Ni) + α (35 Ni)
- At point e',
➤ L(21 Ni) + α (46 Ni) + α (40 Ni) + α (35 Ni) + α (31 Ni)
- At point f',
➤ α (46 Ni) + α (40 Ni) + α (35 Ni) + α (31 Ni)



Phase Diagrams Containing 3-Phase Reactions

- Many binary systems produce phase diagrams more complicated than the isomorphous phase diagrams.
- Five such reactions are defined in Figure .

Eutectic	$L \rightarrow \alpha + \beta$	
Peritectic	$\alpha + L \rightarrow \beta$	
Monotectic	$L_1 \rightarrow L_2 + \alpha$	
Eutectoid	$\gamma \rightarrow \alpha + \beta$	
Peritectoid	$\alpha + \beta \rightarrow \gamma$	

- The eutectic and eutectoid reactions are similar in that they both involve the decomposition of a single phase into two solid phases. The *-oid suffix indicates* that a solid, rather than liquid, phase is decomposing.

Phase Diagrams Containing 3-Phase Reactions

Invariant reactions

Eutectic	$L \rightarrow \alpha + \beta$
Peritectic	$\alpha + L \rightarrow \beta$
Monotectic	$L_1 \rightarrow L_2 + \alpha$
Eutectoid	$\gamma \rightarrow \alpha + \beta$
Peritectoid	$\alpha + \beta \rightarrow \gamma$

- The eutectic, peritectic, and monotectic reactions are part of the ***solidification process***.
- The eutectoid and peritectoid reactions are completely ***solid-state reactions***.
- Each of these three-phase reactions occurs at a fixed temperature and composition. The Gibbs phase rule for a three-phase reaction is (at a constant pressure),

$$1 + C = F + P$$

$$F = 1 + C - P = 1 + 2 - 3 = 0$$

- When the three phases are in equilibrium during the reaction, there are no degrees of freedom. As a result, these reactions are called ***invariant reactions***

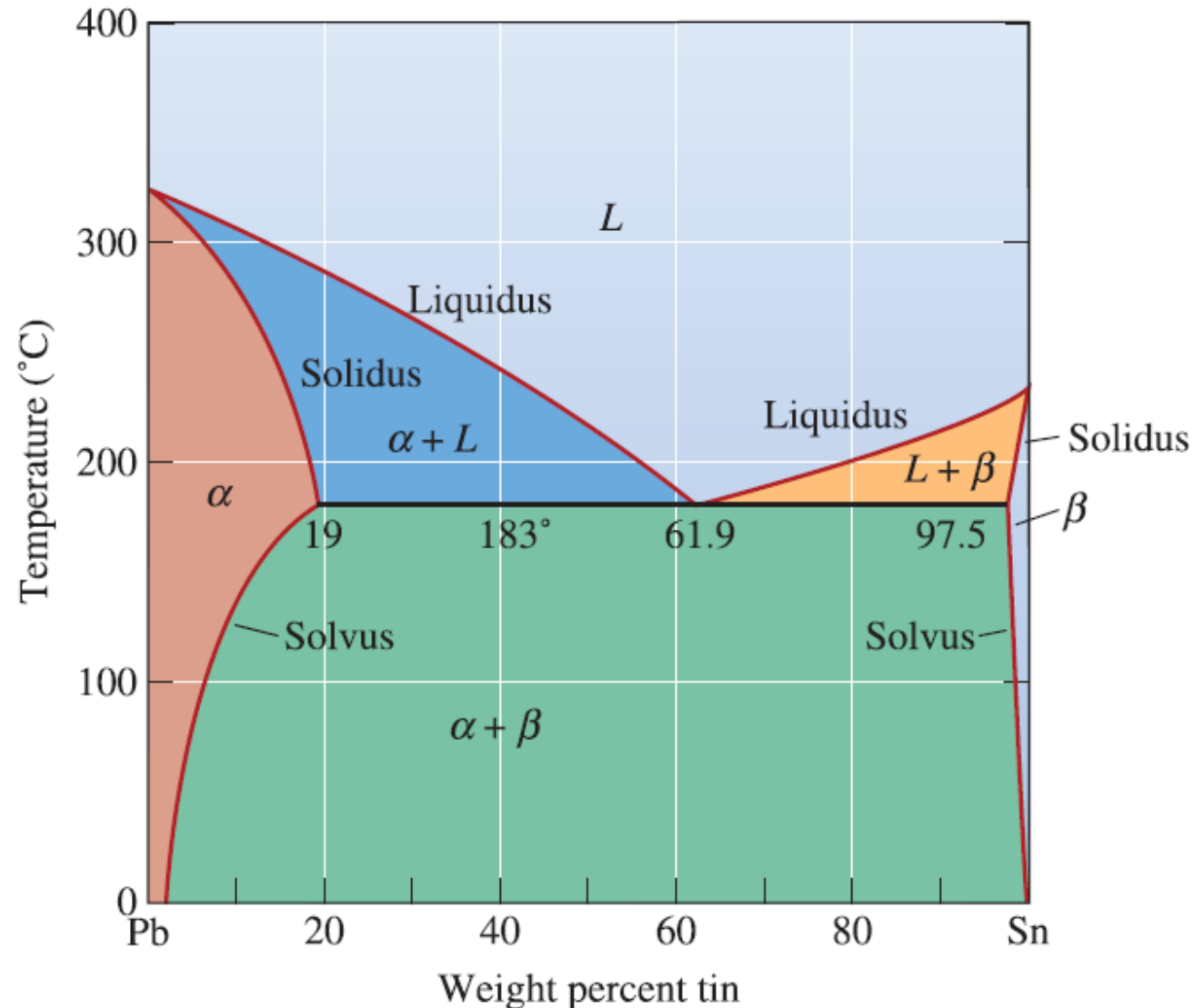
Phase Diagrams Containing 3-Phase Reactions

Procedure to identify each of these reactions in a phase diagram,

- 1) Locate a horizontal line on the phase diagram. The horizontal line, which indicates the presence of a three-phase reaction, represents the temperature at which the reaction occurs under equilibrium conditions.
- 2) Locate three distinct points on the horizontal line: the two endpoints plus a third point, in between the two endpoints of the horizontal line. This third point represents the composition at which the three-phase reaction occurs. In Figure, the point in between has been shown at the center; however, on a real phase diagram, this point is not necessarily at the center.
- 3) Look immediately above the in-between point and identify the phase or phases present; look immediately below the point in between the end points and identify the phase or phases present. Then write the reaction from the phase(s) above the point that are transforming to the phase(s) below the point. Compare this reaction with those in Figure to identify the reaction.

Binary Eutectic alloy system (Lead-Tin System)

- In some binary alloy systems, components have limited solid solubility.
- In the eutectic system between two metals A and B, two solid solutions, one rich in Pb (α) and another rich in Sn (β) form.
- In addition to liquidus and solidus lines there are two more lines on A and B rich ends which define the solubility limits B in A and A in B respectively. These are called ***solvus lines***.



Binary Eutectic alloy system (Lead-Tin System)

Solid-Solution Alloys

- Alloys that contain 0 to 2% Sn behave exactly like the copper-nickel alloys; a single-phase solid solution forms during solidification (Figure 11-7).
- These alloys are strengthened by solid-solution strengthening, strain hardening, and controlling the solidification process to refine the grain structure.

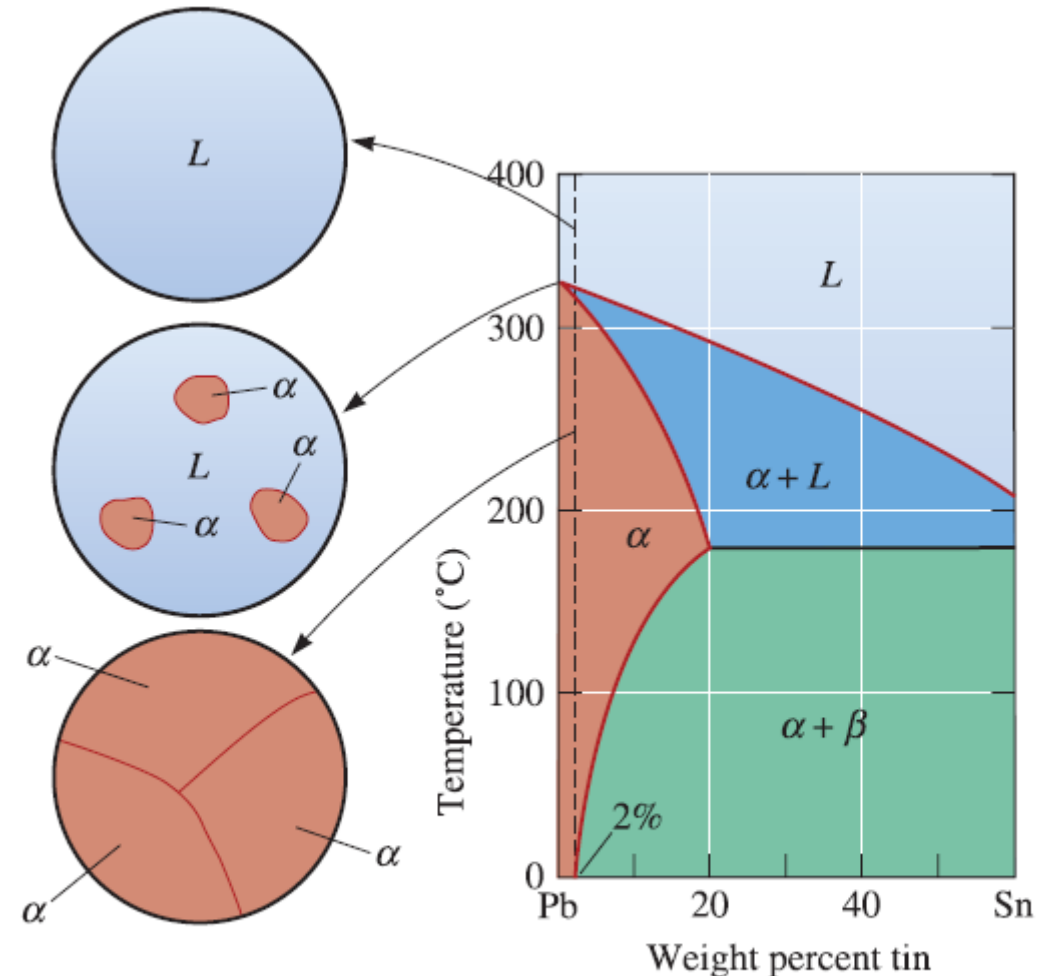


Figure 11-7 Solidification and microstructure of a Pb-2% Sn alloy. The alloy is a single-phase solid solution.

Binary Eutectic alloy system (Lead-Tin System)

Alloys That Exceed the Solubility Limit (between 2% and 19% Sn)

- solidify to produce a single solid solution α .
- As alloy continues to cool, a solid-state reaction occurs, permitting a second solid phase (β) to precipitate from the original phase (α).
- At 0°C, only 2% Sn can dissolve in α . As the temperature increases, more tin dissolves into the lead until, at 183°C. (Solvus line).
- A vertical line on a phase diagram (e.g., Figure 11-8) that shows a specific composition is known as an **isopleth**. Determination of reactions that occur upon the cooling of a particular composition is known as an **isoplethal study**.

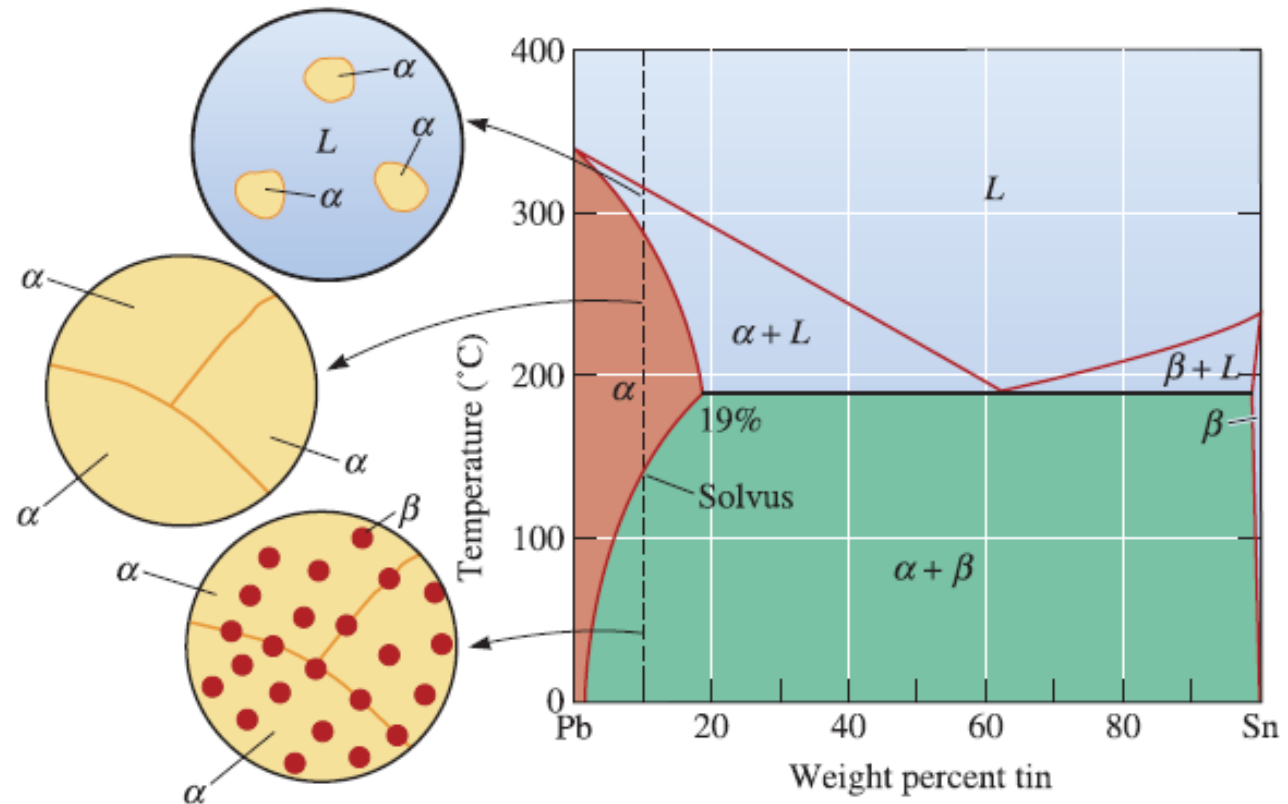


Figure 11-8 Solidification, precipitation, and microstructure of a Pb-10% Sn alloy. Some dispersion strengthening occurs as the β solid precipitates.

Binary Eutectic alloy system (Lead-Tin System)

- alloy containing 61.9% Sn has the eutectic composition. This is the composition for which there is no freezing range (i.e., solidification of this alloy occurs at one temperature, 183°C in the Pb-Sn system).
- Above 183°C, the alloy is all liquid and, therefore, must contain 61.9% Sn. After the liquid cools to 183°C, the eutectic reaction begins
- Two solid solutions α and β are formed.
- compositions of the two solid solutions are given by the ends of the eutectic line

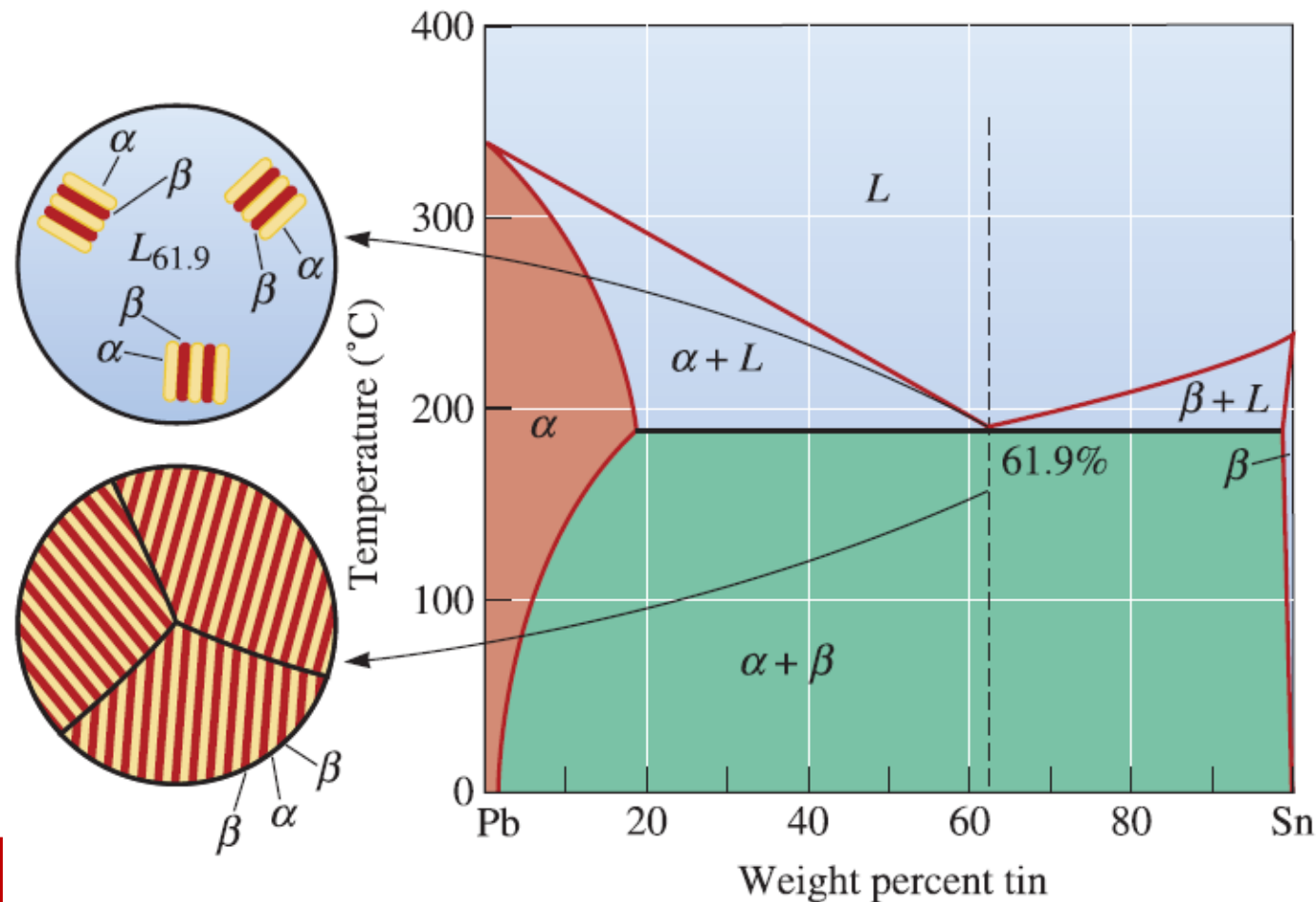
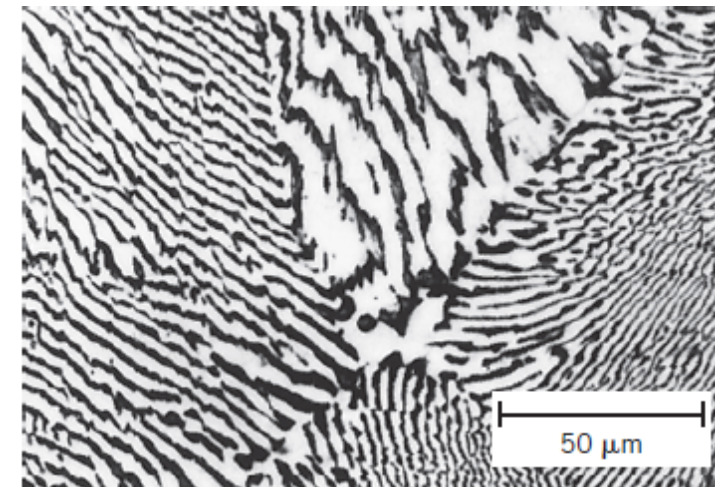
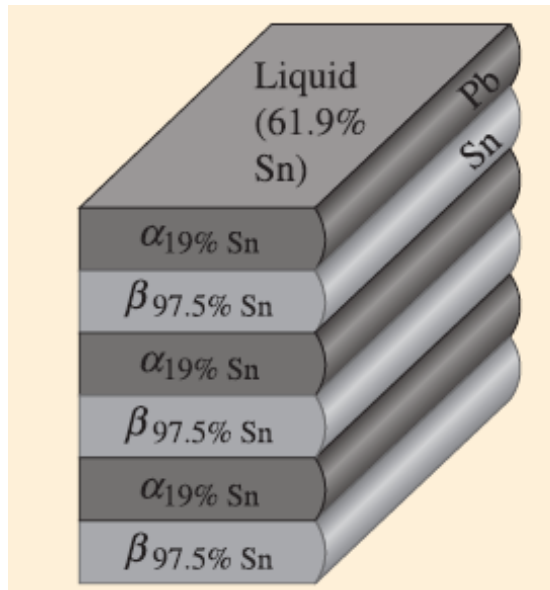


Figure 11-9 Solidification and microstructure of the eutectic alloy Pb-61.9% Sn.

Binary Eutectic alloy system (Lead-Tin System)

- During eutectic solidification, as atoms are redistributed, a characteristic microstructure develops. In the lead-tin system, the solid α and β phases grow from the liquid in a **lamellar, or plate-like**, arrangement.
- The product of the eutectic reaction has a characteristic arrangement of the two solid phases called the **eutectic microconstituent**.
- Atom redistribution during lamellar growth of a lead–tin eutectic. Tin atoms from the liquid preferentially diffuse to the β plates, and lead atoms diffuse to the α plates



microstructure of a lead–tin alloy of eutectic composition consists of alternating layers of a lead rich-phase solid solution (dark layers), and a tin-rich-phase solid solution (light layers).

Binary Eutectic alloy system (Lead-Tin System)

Hypoeutectic and Hypereutectic Alloys

- A ***hypoeutectic alloy*** is an alloy with a composition between that of the left-hand end of the tie line defining the eutectic reaction and the eutectic composition.
- An alloy composition between that of the right-hand end of the tie line defining the eutectic reaction and the eutectic composition is known as a ***hypereutectic alloy***.

Binary Eutectic alloy system (Lead-Tin System)

- The solidification and microstructure of a hypoeutectic alloy (Pb-30% Sn).
- the solid α phase that forms when the liquid cools from the liquidus to the eutectic the ***primary or proeutectic microconstituent***.

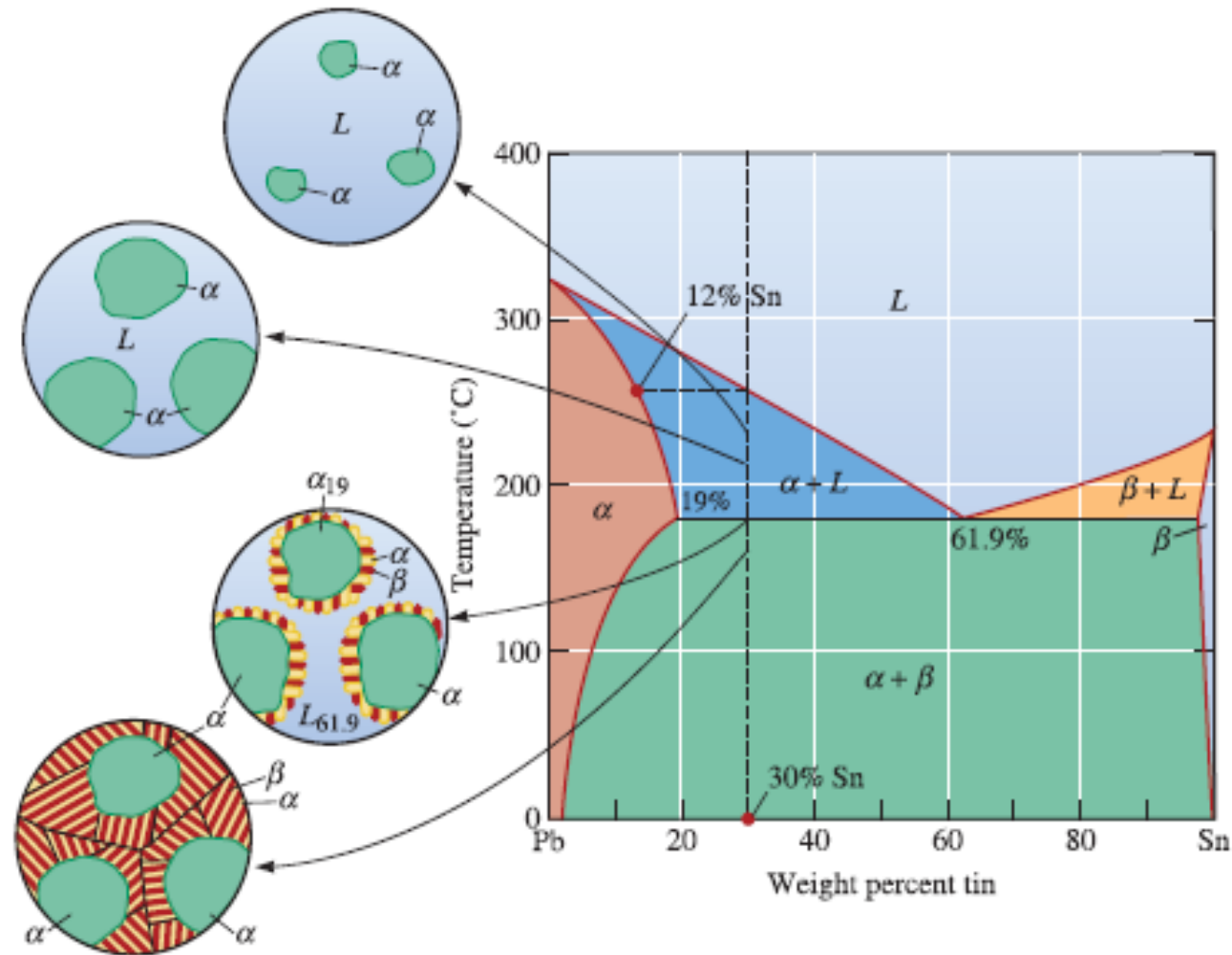


Figure 11-13 The solidification and microstructure of a hypoeutectic alloy (Pb-30% Sn).

Binary Eutectic alloy system (Lead-Tin System)

- The cooling curve for a hypoeutectic alloy is a composite of those for solid-solution alloys and “straight” eutectic alloys (Figure). A change in slope occurs at the liquidus as primary begins to form. Evolution of the latent heat of fusion slows the cooling rate as the solid grows. When the alloy cools to the eutectic temperature, a thermal arrest is produced as the eutectic reaction

proceeds at 183°C.

The solidification sequence is similar in a hypereutectic alloy, giving the microstructure shown in

Figure 11-14(b).

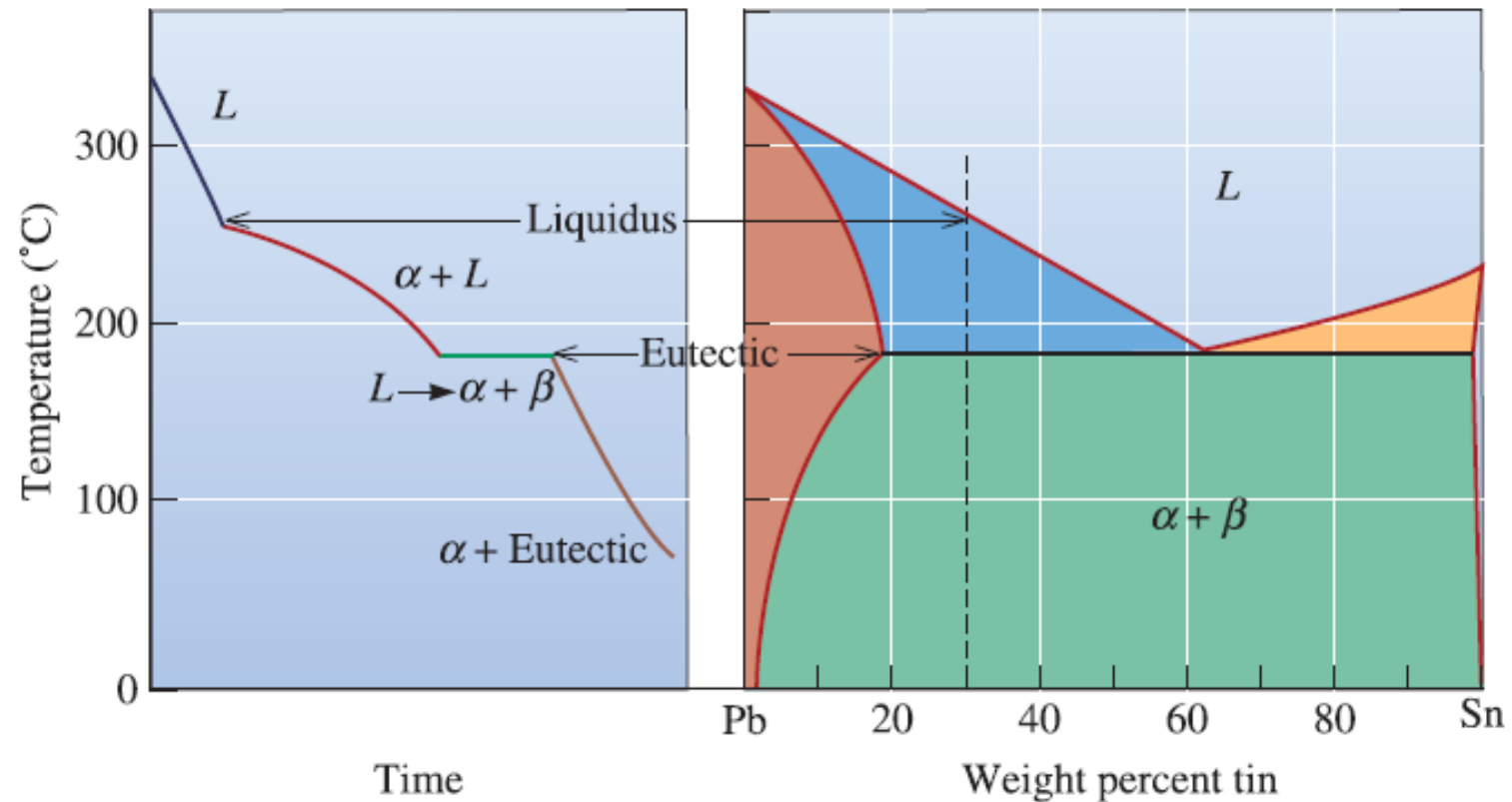


Figure 11-15 The cooling curve for a hypoeutectic Pb-30% Sn alloy.

Binary Eutectic alloy system (Lead-Tin System)

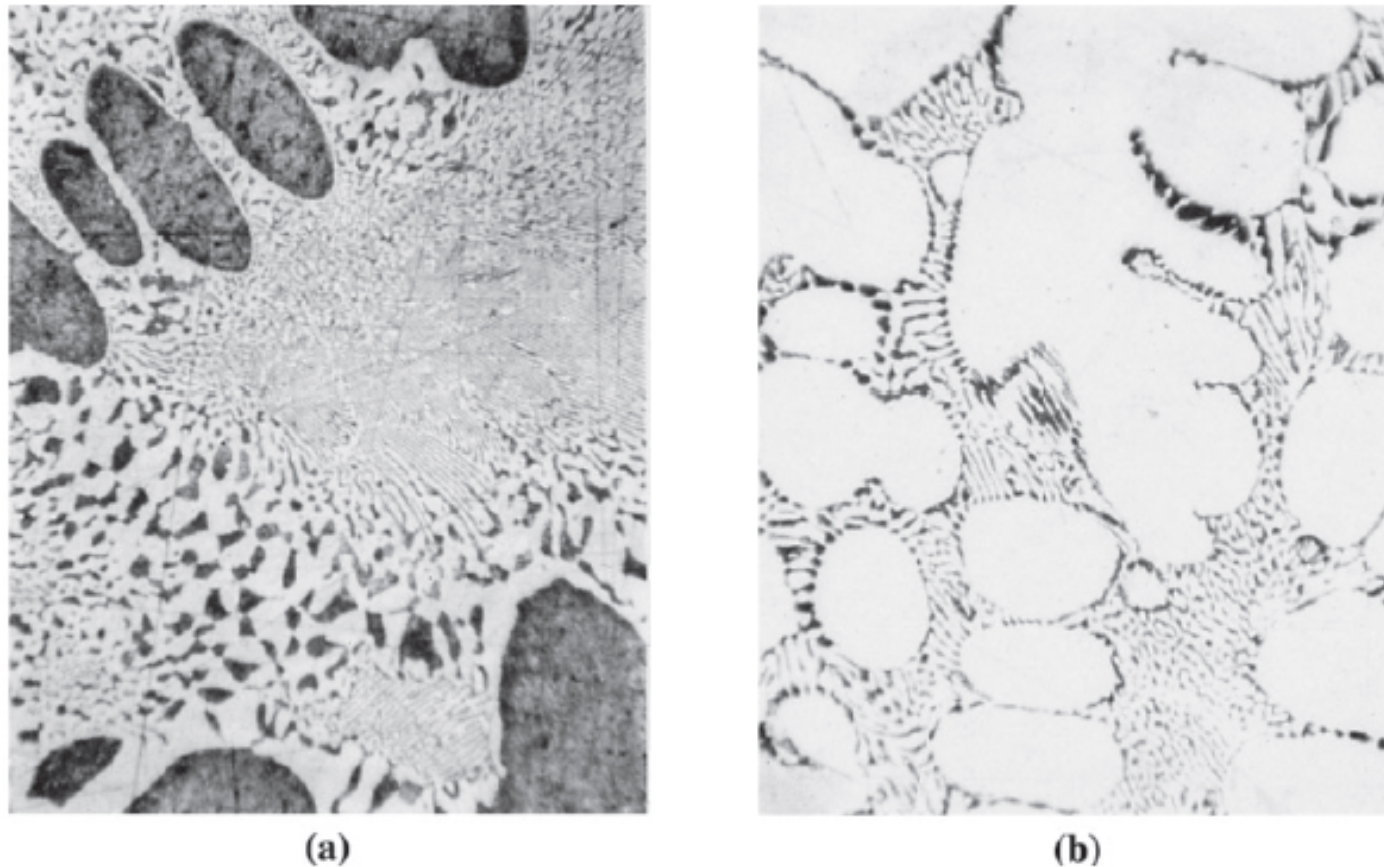
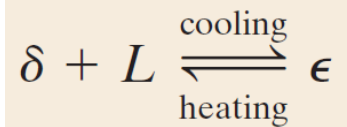


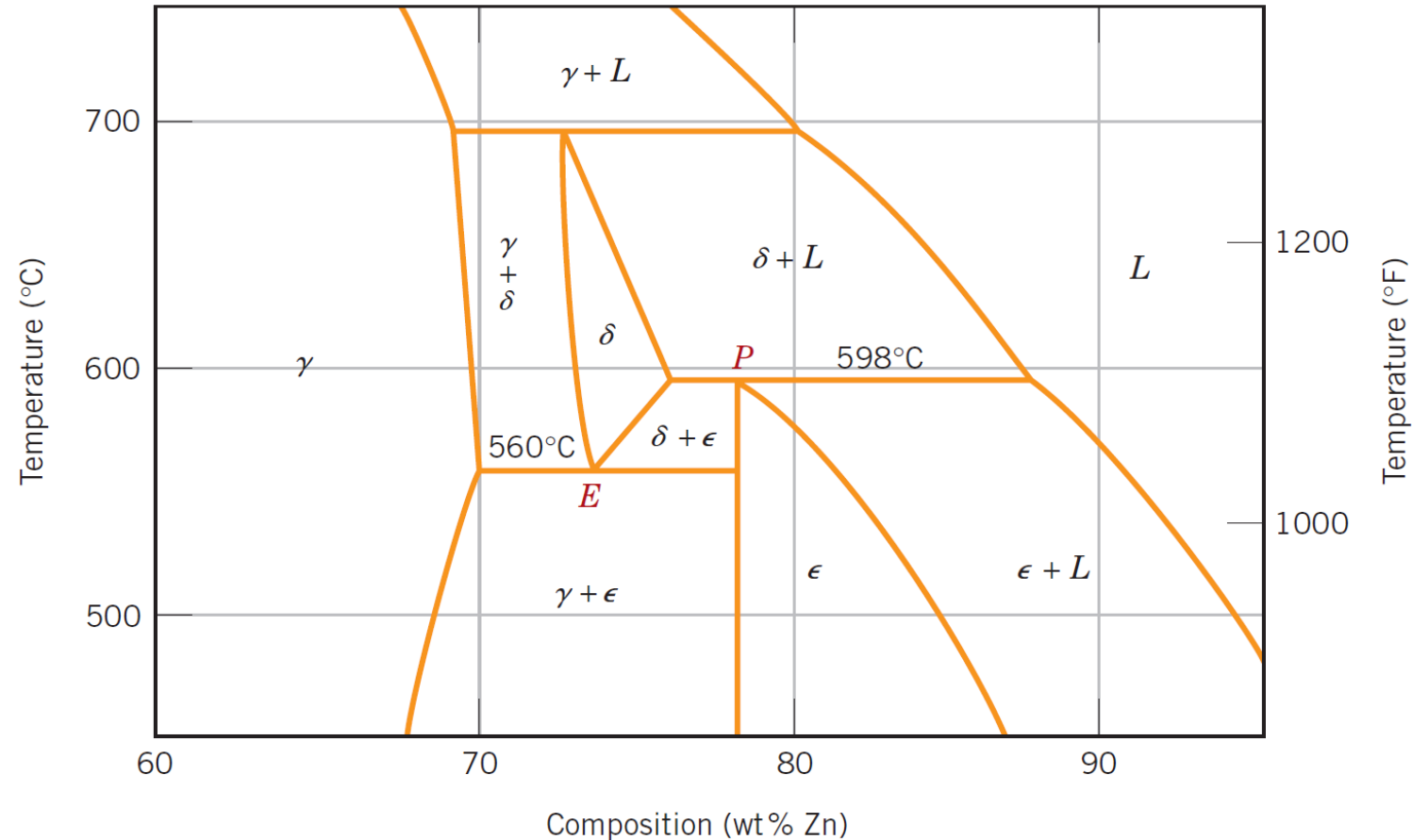
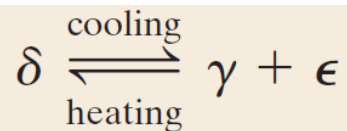
Figure 11-14 (a) A hypoeutectic lead-tin alloy. (b) A hypereutectic lead-tin alloy. The dark constituent is the lead-rich solid α , the light constituent is the tin-rich solid β , and the fine plate structure is the eutectic ($\times 400$). (Micrographs reprinted courtesy of Don Askeland.)

Eutectoid and Peritectic system

- **Peritectic reaction** in which a solid and a liquid combine to produce a second solid on cooling. **(point P).**



- **Eutectoid reaction** in which a solid and a liquid combine to produce a second solid on cooling. **(point E)**



A region of the copper–zinc phase diagram that has been enlarged to show eutectoid and peritectic invariant points, labeled E (560°C, 74 wt% Zn) and P (598°C, 78.6 wt% Zn), respectively.